

Introduction to Applications and beyond IIA [IIA – Lect._____]

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**Computational Intelligence
& Machine Learning Group**

DRAFT, please do not circulate! 2026

ToC

Main parts of this special lecture :

- 1. Programme** of IIA of this year
- 2. General applications and related areas**
- 3. How to study more in AI/ML?**
- 4. Research instances @ Unipi**
 - a. Examples from past projects
 - b. Examples of on-going projects and thesis proposals

Programma IIA PARTE 1:

Problem solving



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Risoluzione di problemi come ricerca:

- Agenti: caratteristiche, razionalità, proprietà degli ambienti, struttura degli agenti
- Formulazione di problemi come ricerca in uno spazio di stati
- Strategie di ricerca non informata: BF, DF, DL, ID, Bidirezionale, UC, ad albero e a grafo, analisi
- Strategia di ricerca informata/euristica: best-first, greedy best first, A*, euristiche ammissibili e consistenti, proprietà, costruzione di euristiche, beam search, IDA*
- Giochi con avversario e cenni a CSP → **Non quest'anno**
- Ricerca locale: steepest ascent/descent, tempra simulata, ricerca local beam, algoritmi genetici, spazi continui (gradiente)
- Azioni non deterministiche e ambiente parzialmente osservabile: Piani condizionali, alberi di ricerca AND-OR (cenni)

Programma IIA PARTE 2:

Logica



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Rappresentazione della conoscenza e ragionamento

- Agenti basati sulla conoscenza: calcolo proposizionale e logica del prim'ordine
- Algoritmi efficienti per la soddisfacibilità: model checking, DPLL, WalkSAT
- Deduzione automatica: trasformazione in forma a clausole, unificazione e metodo di risoluzione
- Sistemi a regole: risoluzione SLD e programmazione logica
- Rappresentazioni strutturate (frames, semantic networks) → **Non quest'anno**

Programma IIA PARTE 3:

Machine Learning (en)



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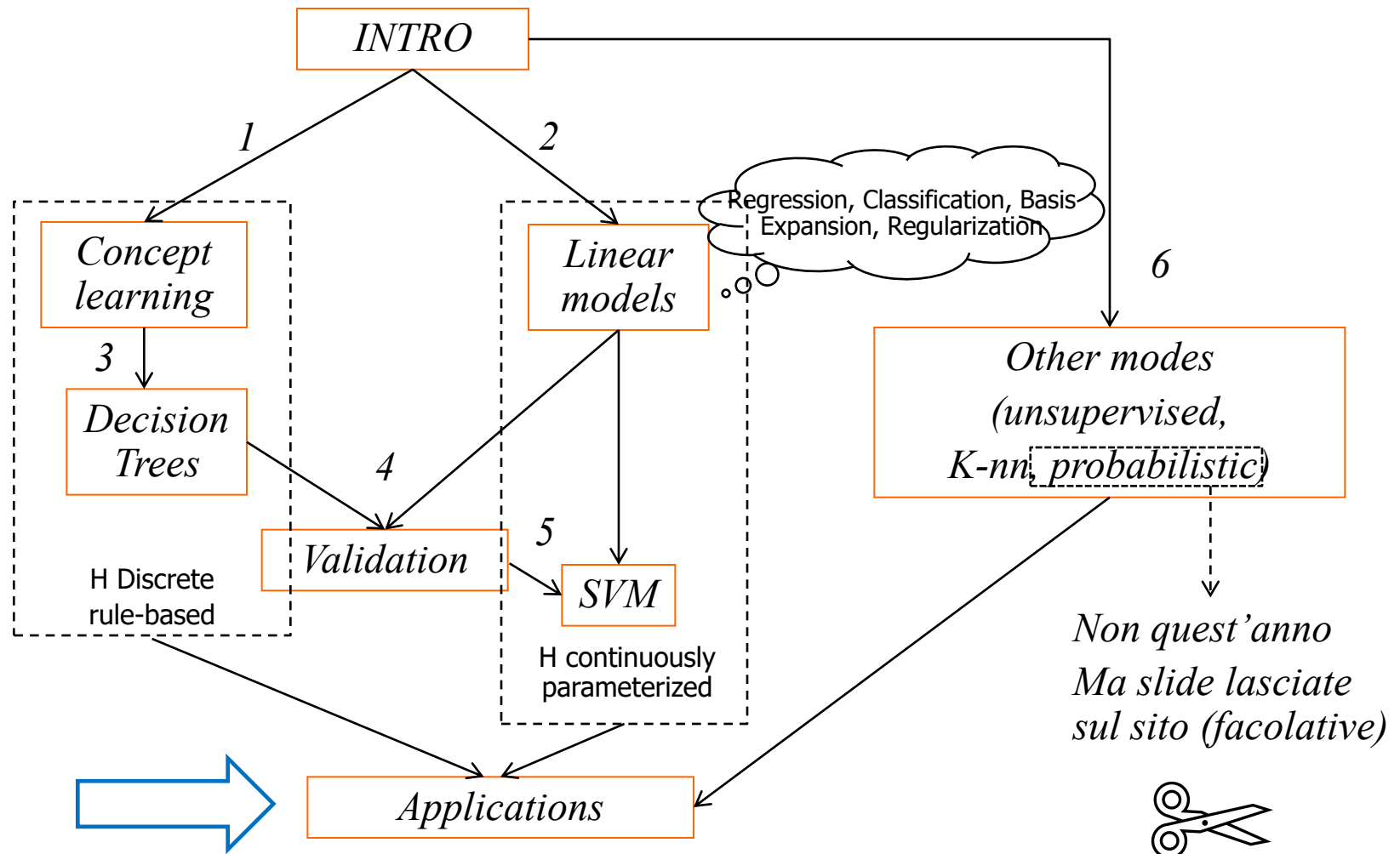
- Introduction: role of ML, learning as function approximation, supervised and unsupervised tasks, generalization
- Introduction to Regression and Classification with Linear models,
 - Linear basis expansion and regularization
- Introduction to Discrete rule-based models:
 - Concept learning: Rule based hypothesis space and Inductive Bias
 - Introduction to Decision Trees learning
- Elements of Validation techniques and Statistical Learning Theory framework
- Introduction to the use of SVM: margin, soft-margin, Kernel, grid-search
- Instance Based (K-nearest-neighbors)
- Unsupervised Learning: k-means clustering
- Other tasks and approaches in ML: short survey and neural networks ideas
- [Quantify Uncertainty in AI: probability, Bayesian networks and] → NOT this year
- Naïve Bayes. Naïve Bayes Learning → NOT this year
- Applicative examples

ML part

Course flow



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La linea (parte ML)

- **Introduzione al ML** attraverso diversi “linguaggi” partendo in parallelo dallo sviluppo dei modelli nel continuo e discreto con forti assunzioni nello spazio ipotesi (H)
 - verso **modelli flessibili** continui e discreti,
 - aprendo al concetto di teorico di complessità e pratico di validazione dei modelli per il **controllo della flessibilità** (trade-off underfitting/overfitting).
 - Complessivamente coperti esempi di **modelli** dei paradigmi symbolic, sub-symbolic, instance based (~~and Bayesian~~)
- 2 concetti finali per **unificare**
 - Learning as **inferring** hp over data (from examples and by inductive bias)
 - **Generalization capability** (quality/performance measure in ML) and control of model complexity
 - Flexibility is needed and useful but **complexity control** is needed as well
 - **Validation** is essential (using different data sets in a rigorous way)

APPLICATIONS



Repetita: Machine Learning: When?

- Models to solve *real-world problems* that are difficult to treat with traditional techniques (complementary to analytical models based on previous knowledge, algorithms and imperative programming, classical AI, ...)
- Examples of Appropriate Applications versus Standard Programming:
 - Too difficult (to be formalized by 'hand-made' algorithm)
 - e.g. face recognition: humans can do it but cannot describe how they do it :
 - e.g. voice automatic telephone answering service
 - Not enough human knowledge (e.g., predicting binding strength of molecules to proteins)
 - Personalized behavior (e.g., scoring email messages or web pages according to user preferences)
 - Individualized (intelligent) human-computer interfaces

Applications (examples)



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Very broad since the 80s especially in Robotics, Pattern recognition, Knowledge Discovery and Data Mining.

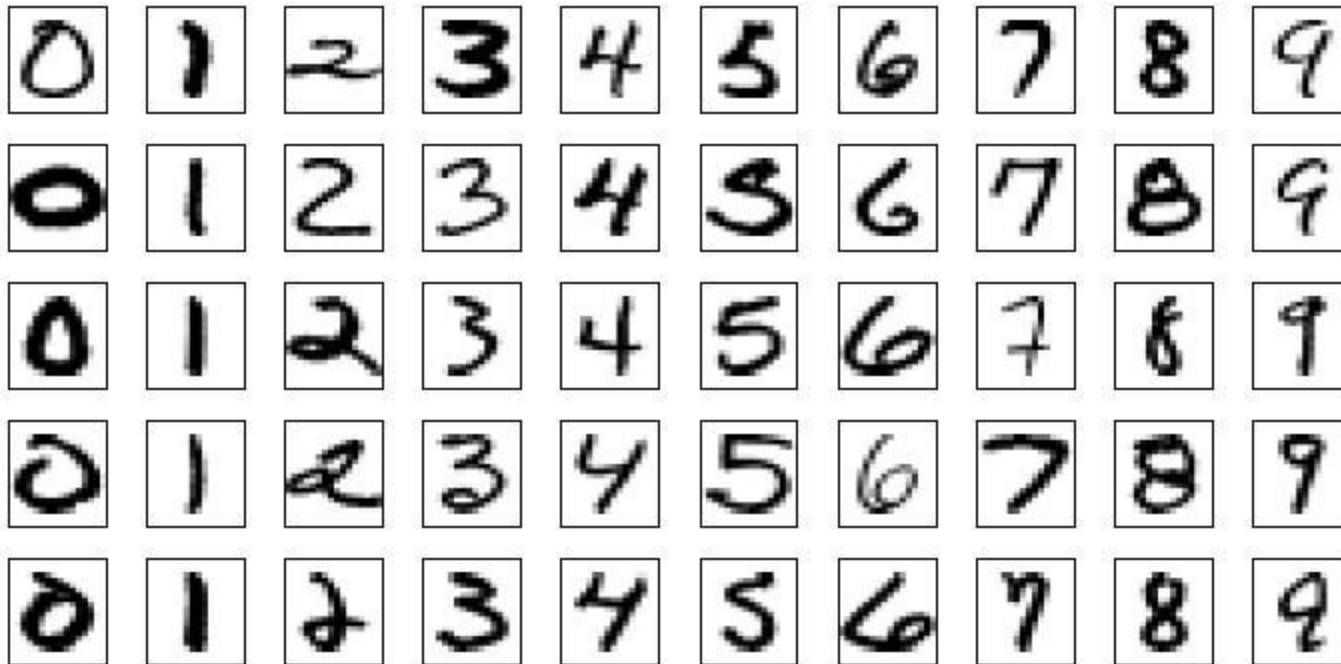
Examples (classic applications)

- Signal and image analysis (eg medical diagnosis)
- Bioinformatics
- Cheminformatics (new material and drug design)
- Computer vision
- Natural Language processing: written and spoken language recognition
- Document processing: Web Doc, e.g. *email spam classification*
- Financial forecasting
- Dynamic processes and control
- ...many many other (see **Intro lecture**)

Specific Case Study (early applications)

- Handwritten recognition
 - Automatic mail sorting by postal code
 - Checks control
 - OCR
 - ...
- NIST (National Institute of Standards and Technology – USA)
benchmark data set: MNIST
- Nice track of ML application evolution, e.g.:
 - K-NN, linear classifier, K-NN with specialized distance (!), Neural Networks, special NNs, SVM, committee/ensemble of models approaches (up to 2012),... *nowadays* deep NN, ...

Character recognition: classification



Examples of training cases from ZIP code data.

Each image is a (normalized) 16×16 8-bit grayscale representation of a handwritten digit (Le Cun et al. 1990).

© Hastie, Tibshirani & Friedman 2009

MNIST Handwritten Digit Recognition Benchmark



- 60000 image for training 10000 for test.
 - Initial misclassification rate of 5%/20% (many errors due to lack of invariance as rotations etc.)
 - Humans 0.2%/2.5%
- Progress of the results (up to 2012) at <http://yann.lecun.com/exdb/mnist/> (nowadays out of service?)
- https://en.wikipedia.org/wiki/MNIST_database
- Progress by Neural Networks (NN) Convolutional NN (special architecture to take in account local invariant feature), SVM, Deep Learning (NN with many hidden layers):
 - Recent methods pre-train each hidden CNN layer one by one in an unsupervised fashion then use supervised learning to achieve **0.39%** (2006)/**0.35%** (2011)
- Best 2012: 0.23% error (committee of CNN)
2016: **0.21%**, ... and even better recently,
but ... is this still a valid "test set"?

Go & Machine Learning



Go winner
(DeepMind - Big G)



Nature 529, 445–446 (28 January 2016)
*The news: Deep-learning software defeats
human professional for the first time.*

Nature 529, 484–489 (il metodo)
*Mastering the game of Go with deep
neural networks and tree search*

Final results on March 2016 competition:
AlphaGo 4 - Lee Sedol 1.

<https://deepmind.com/alpha-go>

Aggiornamenti continui di
automazione e nuove vittorie...

E applicazioni al “mondo reale”...



Self-driving cars

Abstract (Nature 529, 484–489)

- The game of Go has long been viewed as the most challenging of classic games for artificial intelligence owing to its **enormous search space** and the **difficulty of evaluating board positions and moves**. Here we introduce a new approach to computer Go that **uses 'value networks' to evaluate board positions and 'policy networks' to select moves**. These **deep neural networks** are trained by a novel combination of supervised learning from human expert games, and reinforcement learning from games of self-play.
- Without any lookahead search, the neural networks play Go at the level of state-of-the-art Monte Carlo tree search programs that simulate thousands of random games of self-play.
- We also introduce a **new search algorithm (*)** that combines Monte Carlo simulation with value and policy networks. Using this search algorithm, our program AlphaGo achieved a 99.8% winning rate against other Go programs, and defeated the human European Go champion by 5 games to 0.
- This is the first time that a computer program has defeated a human professional player in the full-sized game of Go, a feat previously thought to be at least a decade away.

() Related to minimax
& alpha-beta*

General challenges

- Build autonomous Intelligent/learning machines:
 - Robotics, HRI, search engines, ...
- Build powerful tools for emerging challenges in intelligent data analysis
 - Tools for the “data scientist”
- Open new areas of applications in CS: innovative interdisciplinary open problems
 - Fantasy is your limitation !
 - ML in the era of “changing of paradigm in science, in which scientific advances are becoming more and more *data-driven*”

Challenges by Source of data

2000's

- **Bio-Cheminformatics** (*-omics, personalized medicine, toxicology)
- **Web** data

Recent hot topics ("buzzwords"?)

- **Smart** systems (design smart things and make human life better): from *Smart Cities* to Ambient Assisted Living to the Internet of Things (One trillion = 10^{12} devices with sensors of all kinds are expected to be connected to the Internet by 2022 (IEEE news March 2014))
- **Social, Big data** : e.g. social network data, or again IoT

Growing data sources opens up a huge application area for ML and related areas (also in terms of employment opportunities)

Related area:

Pattern Recognition

- Assignment of a label to a given input value/signal: classification in ML
- Using ML methods is often largely overlapped to it for means and aims.
- Match ML motivations: Difficult to **formalize** exactly the solution of the problem: Possible presence of **noise** and **ambiguous** data; see "ML: when" slide in the intro.

ALSO (not here):

- *Medical images*
- *Patterns in general signals*
- ...



Examples of patterns: sound wave, fingerprint, trees, face, bar code, and character images.

Related area:

Data Mining and KDD

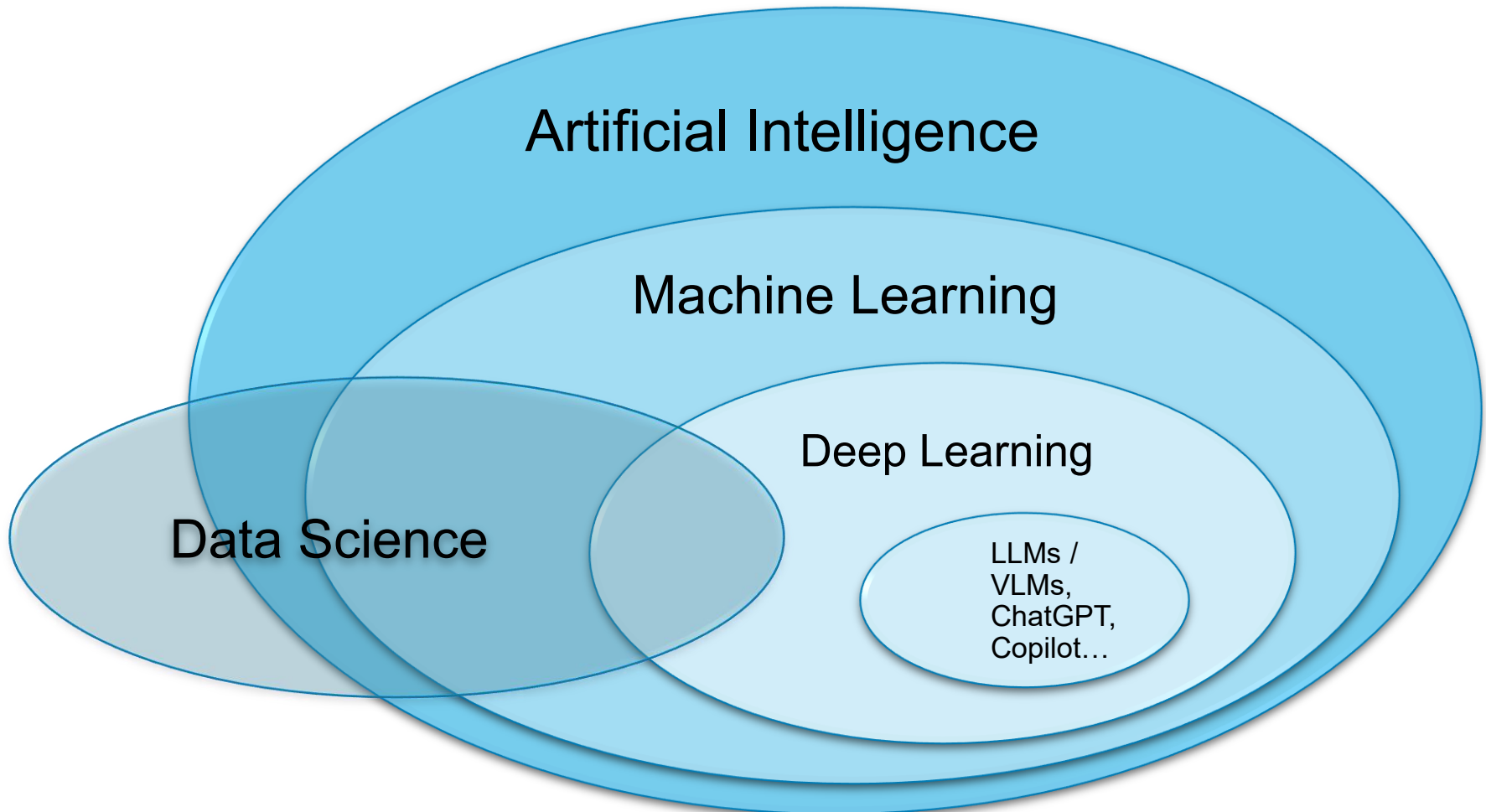


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- More recent field for data analysis (from ML and databases area)
- The (**KDD**) Knowledge Discovery in (DB) process (cycle) include also the phases of
 - (pre) data management and preprocessing
 - (post) extraction/interpretation/evaluation of information
 - privacy/ethics aspects
- **DM**: emphasis on descriptive analysis of data including clustering, frequent/unusual pattern discovery, associative rules
 - Subject: learning methods (ML) versus data/pattern (DM)
- Also applies ML results/tools (e.g. for classification purposes) within the aim of discovering *patterns* in **large databases**
- Efficiency and interpretability are key points
- Applications on large databases : marketing (e.g. market basket analysis), financial (e.g. fraud detection), epidemiologic data, ...
- Nowadays: "*Big Data*", "**Data Science**", ...

AI/ML/DL/DS...

Not all the story but nice view at a glance



A more general area:

Computational Intelligence



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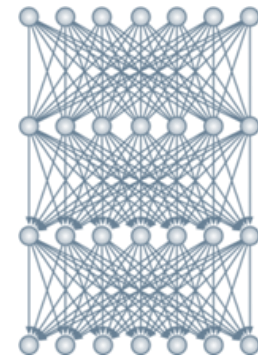
- Relevance of the treatment of uncertainty and noisy information for AI real complex problems to deserve an autonomous field name: **Computational Intelligence**
- Often nature-inspired computational methodologies, including:
 - artificial neural networks,
 - evolutionary computation
 - fuzzy logic.
- Also swarm intelligence, artificial immune systems, ...
- Related to **Soft computing**: tolerance of imprecision, uncertainty, partial truth, and approximation.

Other related fields

- *Bioinformatics*
- *Natural Language Processing* (from Machine translation to speech recognition): [AIMA chap 23 & 24](#)
- *Computer Vision*: [AIMA chap 25](#)
- *Perception & Robotics*: [AIMA chap 26](#)
- *Sensor networks* → Intelligent Sensor Networks (with adaptive properties)
- *Information Retrieval*
- ...

DL impact

- And Deep Learning (DL) is revolutionizing the AI approach to these fields (see previous lecture)
- ML: Changing how to get knowledge into a computer: from hard-coded to extracting them directly from data
- Neural Networks provide an approach to Representation Learning (RL)
- DL introduce a hierarchical approach to RL combining simple concept to abstract concepts (learning multi level of composition)
- Data + HPC + modern ML/DL → New AI
- New AI → new society ?!?!?



State of AI Performance – AI-index Report 2024



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Select AI Index technical performance benchmarks vs. human performance

Source: AI Index, 2024 | Chart: 2024 AI Index report

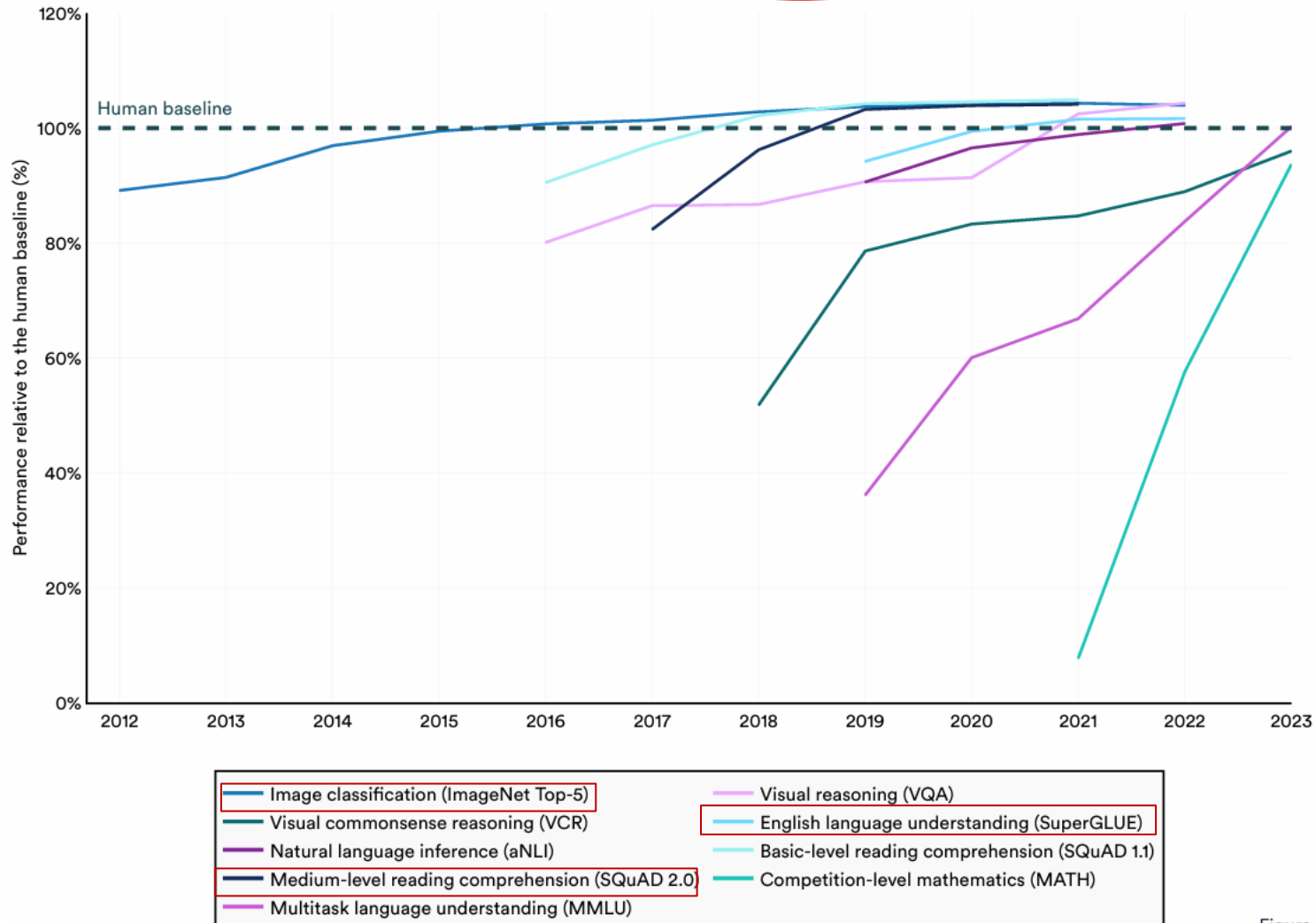


Figure 2.1.16²

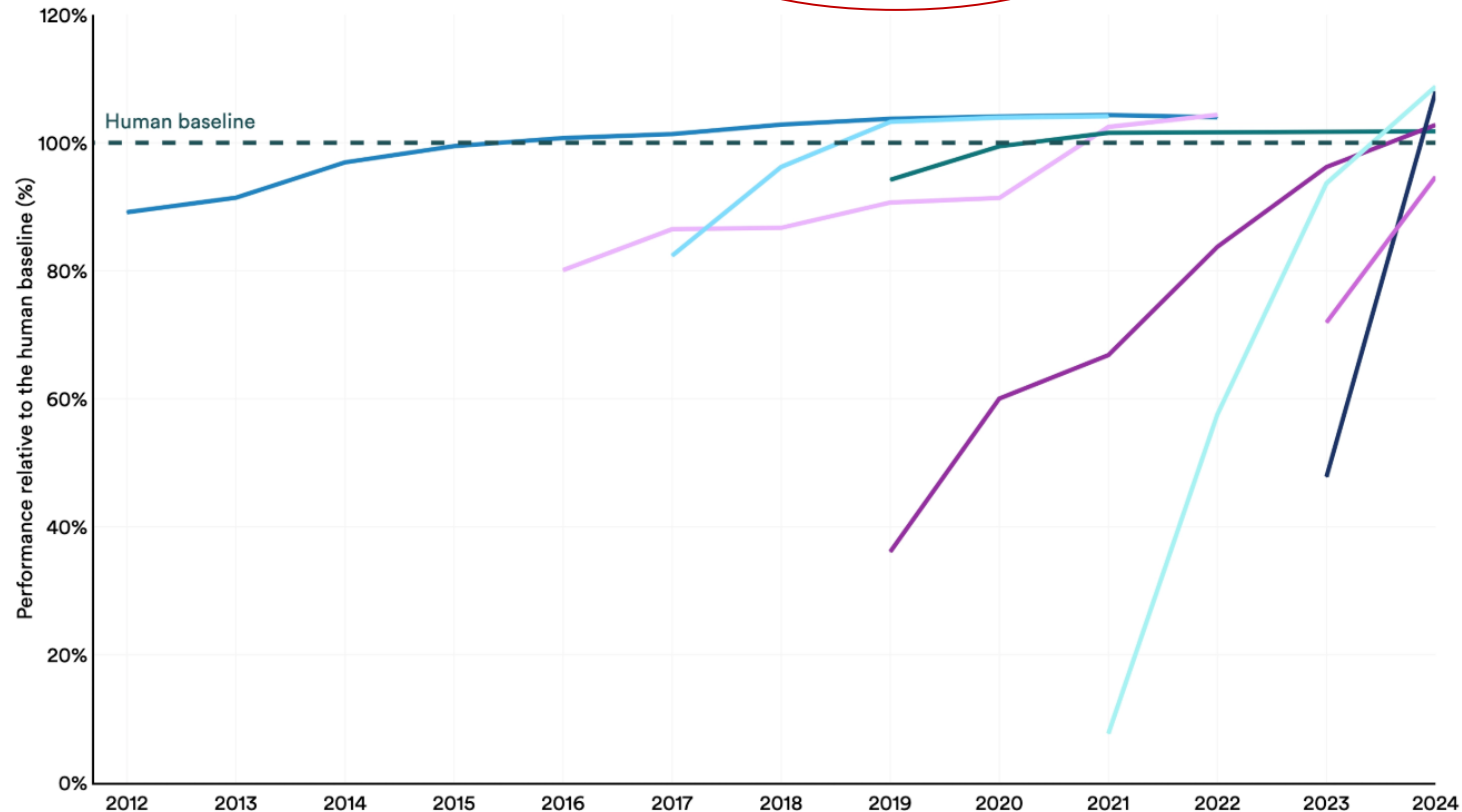
State of AI Performance – AI-index Report 2025



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Select AI Index technical performance benchmarks vs. human performance

Source: AI Index, 2025 | Chart: 2025 AI Index report



Turing award 2018

- On 27-th March 2019, the Association for Computing Machinery, the world's largest society of computing professionals, announced that Drs. Hinton, LeCun and Bengio had won this year's Turing Award for their work on **neural networks**. The Turing Award, which was introduced in 1966, is often called the Nobel Prize of computing, and it includes a \$1 million prize, which the three scientists will share.
- "For conceptual and engineering breakthroughs that have made deep neural networks a critical component of computing"



Y. Bengio



G. Hinton



Y. LeCun

Nobel Prize – Physics 2024

“for foundational discoveries and inventions that enable machine learning with artificial neural networks”



John Hopfield
Born: 1933, USA



Geoffrey E. Hinton
Born: 1947, UK

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- https://www.nobelprize.org/uploads/2024/11/Slideshow_Physics_2024_NobelPrizeLessons.pdf

Nobel Prize – Chemistry 2024

David Baker is awarded "*for computational protein design*" and Demis Hassabis and John Jumper "*for protein structure prediction*".



David Baker
Born: 1962, USA



Demis Hassabis
Born: 1976, UK



John Jumper
Born: 1985, USA

- “I’ve always thought if we could build AI in the right way, it could be the ultimate tool to help scientists, help us explore the universe around us.
Demis Hassabis (Alpha Fold - Deep Mind)
- https://www.nobelprize.org/uploads/2024/10/Slideshow_Chemistry_2024_NobelPrizeLessons.pdf

Oltre IIA

- IIA = Introduzione all'intelligenza artificiale:
 - lungi da coprire la disciplina dell'AI
 - vuol essere un presentazione rigorosa ma resta necessariamente estremamente parziale
- Ma l'obiettivo ultimo è un altro:
- Plutarco: *la mente non è un vaso da riempire ma un fuoco da accendere*
 - *Didattica*
 - *Ricerca*



How to study more in AI/ML?



Master Degree – Pisa:

- **Computer Science**
- **Data Science and Business Informatics**
- **Informatics for Digital Health**

Master Degree in Computer Science (Pisa)

- Since 2017 the curricula are:
 - Artificial Intelligence
 - Big Data Technologies
 - ICT Solutions Architect
 - Foundation of Software
- Advantages:
 - Opportunity to specialize in a field (identify your interest, professional qualification (supplementary diploma)), ...to enjoy!
 - Methodological courses for the area at the beginning
 - Show that 2 years more of study can be useful for your future.

AI & Master degree (Pisa)

- AI is related with all the 4 curricula (see the previous discussion on the applications)
- See *characterizing* and *electives* courses in each curricula*
- Of course the AI curriculum is dedicated to *AI* ;-)

Artificial Intelligence



Further motivations

- CS/AI is taking over the world... but *intelligence* is **not** everywhere **yet**... especially for *machines*
 - With more *intelligence* we can address new challenges in a better way
- We (the humanity) need **your** contribution to make it!!!!
- But AI is *not a new topic*, there are already many previous approaches and scientific results, starting at the early stage of CS, that you *need to study* before you can contribute!
- And, thanks to flexibility of ML approaches, it already provides a set of revolutionary approaches to solve complex problems
- Career opportunities range from major companies, to research institutes and academia, and the startup!*
- But they ask a SOLID background in AI/ML to create new solutions!!!

AI curriculum - Plan



- Methodological **characterizing** basis (to build adaptive/intelligent systems):
 - Artificial intelligence fundamentals - 6 CFU
 - Computational mathematics for learning and data analysis - 9 CFU
 - Machine learning - 9 CFU
 - Parallel and distributed systems: paradigms and models - 9 CFU

Blu: shared with other curricula

Red: as characterizing course is only in AI (but can be shared as electives)



AI curricula - Plan

Many other **characterizing** and related fields:

- Generative and deep learning - 9 CFU
- Human language technologies - 9 CFU
- Computer vision - 9 CFU

Group: AI electives (9 CFU)

- Algorithm engineering (KD)
- Data mining (KD)
- Mobile and cyber-physical systems (ICT)
- Algorithm design
- Digital Health lab

Group: AI electives (6 CFU) → new current complete list [here](#), e.g.:

- Information retrieval (BDT)
- Computational neuroscience (ING)
- New: Learning on Graphs (not active yet)
- Social and ethical issues in computer technology
- Robotics
- Semantic web

Further info

Master Degree:

- <https://www.di.unipi.it/en/education/mcs>

AI:

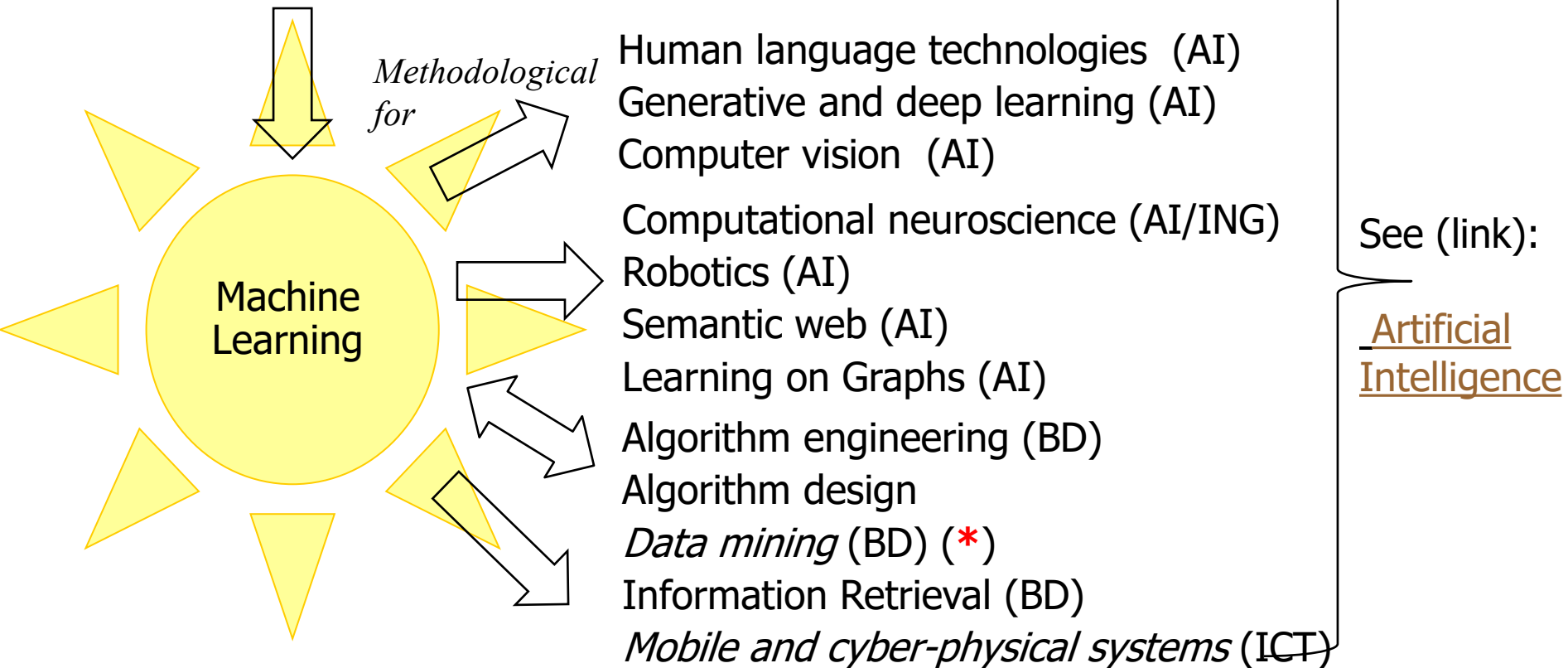
- <https://www.di.unipi.it/en/education/mcs/curricula-en/ai>

ML: Objectives at a Glance

- Introduce the principles and the main paradigms (models and algorithms) for learning from data sets and its applications.
- Provides the methodological foundations for two aims:
 - building new adaptive intelligent systems
 - building powerful predictive models for intelligent data analysis
- METHOD
 - Concepts are introduced progressively, from the simplest approaches (with no prerequisite courses*) up to state-of-the-art models
 - Include **Neural Networks** (up to deep learning), SVM, etc. (which are excluded or mentioned in this course) as well as further study in cases already encountered
 - Emphasis on aspects of
 - General and modern conceptual framing of machine learning
 - a systematic approach based on learning functions from examples
 - principles underlying learning that are common across different methods
 - Critical use (versus limitations) of models for real-world problems
 - Experimental evaluation
 - **PRJ** (try and enjoy)

ML & Intelligent Systems area (New, since a.y. 25/26)

Computational mathematics for learning and data analysis
Parallel and distributed systems: paradigms and models



But also others:* Bioinformatics (BD) Advanced databases (BD)
Big data analytics (WBI) Digital Health lab ...

Further Info?



Please ask

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Research

ML: continuously growing area

- In the modeling-theoretical and applied aspects (e.g. frontier of scientific research)
- See examples in the following

1. CIML group @ UNIFI

2. Research by instances

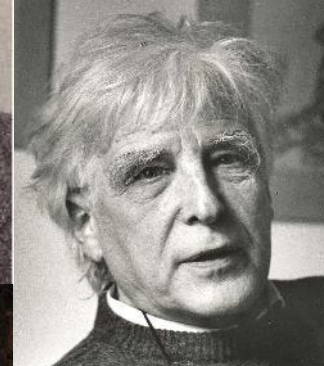


**Computational Intelligence &
Machine Learning Group**

<https://ciml.di.unipi.it/>

Origins

- **Edoardo Renato Caianello** (Napoli, Italy), 60's: Neural networks, Cybernetics
- **Valentino von Braitenberg**: Biocybernetics
- **Antonina Starita**: Pisa "Neurolab", 80's: Computational Intelligence and Bioengineering
- **Alessandro Sperduti** (Pisa & Padova), 90's
- Me ;-)



CIML Group - People

- **Faculty** (9): Prof. A. Micheli (coordinator), Prof. D. Bacciu, Prof. C. Gallicchio, Dr. A. Carta, Dr. A. Ceni, Dr. A. Cossu, Dr. L. Passaro, Dr. M. Podda, Dr. D. Tortorella
- 4+ post-doc researchers
- **12/14+ Phd students (the young energy!)**
- More than **25 years** of research in ML/AI
- Part of the cairne.eu Research Network
(Confederation of Laboratories for Artificial Intelligence Research in Europe)

CAIRNE



**Computational Intelligence &
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<https://ciml.di.unipi.it/>

CIML Group - People

Many PhD students ...



New calls each year, for

- PhD positions in **Computer Science** or
- the **National** PhD in **Artificial Intelligence**



*“tie a knot in
your
handkerchief”*



**Computational Intelligence &
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<https://ciml.di.unipi.it/>

CIML Projects

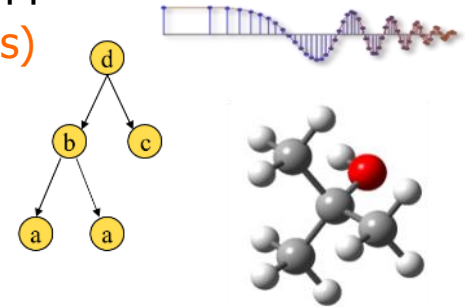
- Some examples of projects @ **CIML-Pisa** 
- On request + Motivational: ***University: didactics & research***



CIML Main areas (overview)

Development of **basic and applied research on Machine Learning**

- **Learning in Structured Domains (SD):** sequence, trees and graphs/networks
 - Pioneering since the **90's** the development of *Recursive Neural Networks*
 - **New models** by **Reservoir Computing**, **Kernels** and **Generative** approaches
 - Leadership in **efficient deep learning** for **structured data (graphs)**
- **Vertical approach to ML:**
 - Theoretical analysis
 - New models development (core!)
 - Applications
- Interdisciplinary applications (med., bio, chem., eng., robotics, ...)
- European projects, see later
- Industrial collaborations



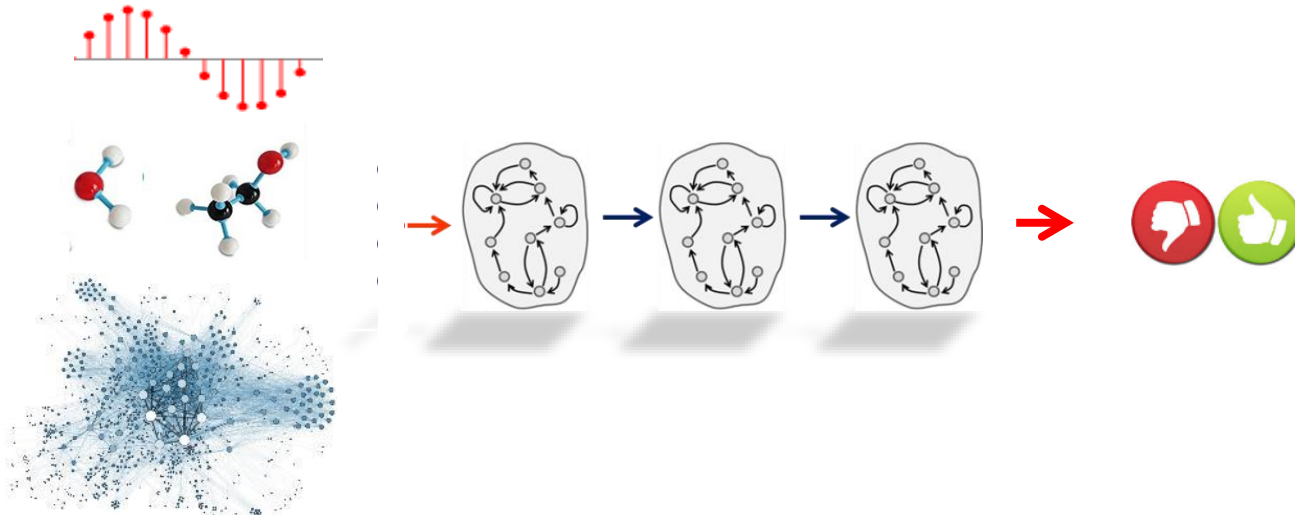
Contact: micheli@di.unipi.it

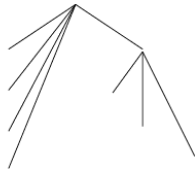
**Computational Intelligence
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Development of **basic and applied research on Machine Learning**

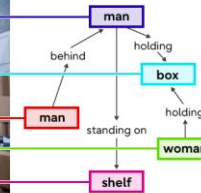
Learning in Structured Domains (SD):

- **Aim: to build efficient learning systems able to process directly complex data**
- *"Because data have relationships"*
- New models for sequences and graphs/networks in the deep neural networks, kernel based-probabilistic and efficient reservoir computing paradigms

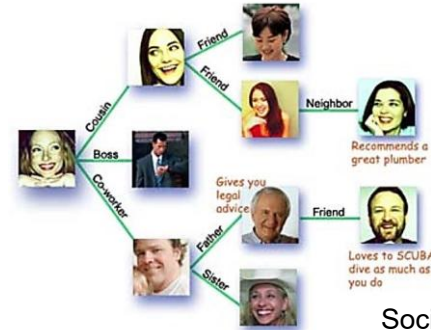




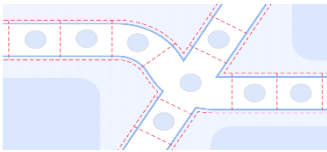
Abstraction of images



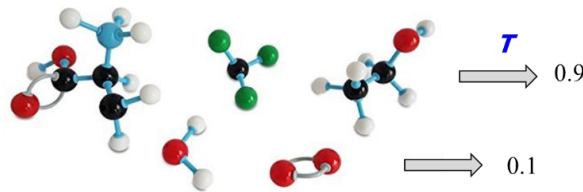
Scene understanding networks



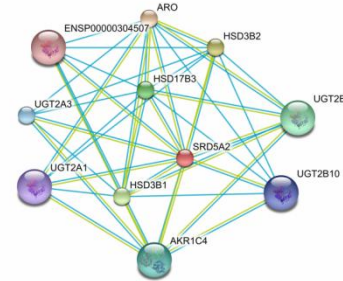
Social networks



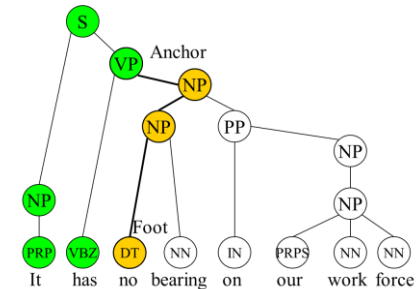
Transportation networks
(traffic predictions on G. maps
@ DeepMind)



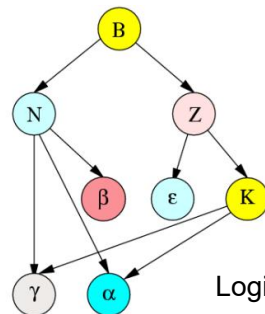
Small molecules



Biological networks
(proteins)

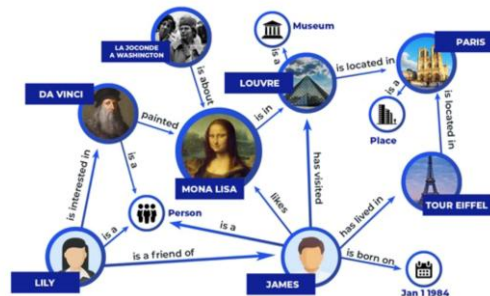


Language parsing

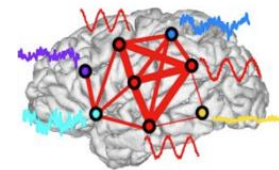


Logic terms

$$B(N(\gamma, \alpha, \beta), Z(\varepsilon, K(\gamma, \alpha)))$$



Knowledge graphs



Brain (networks) modeling

Deep Graph Networks: Evolution

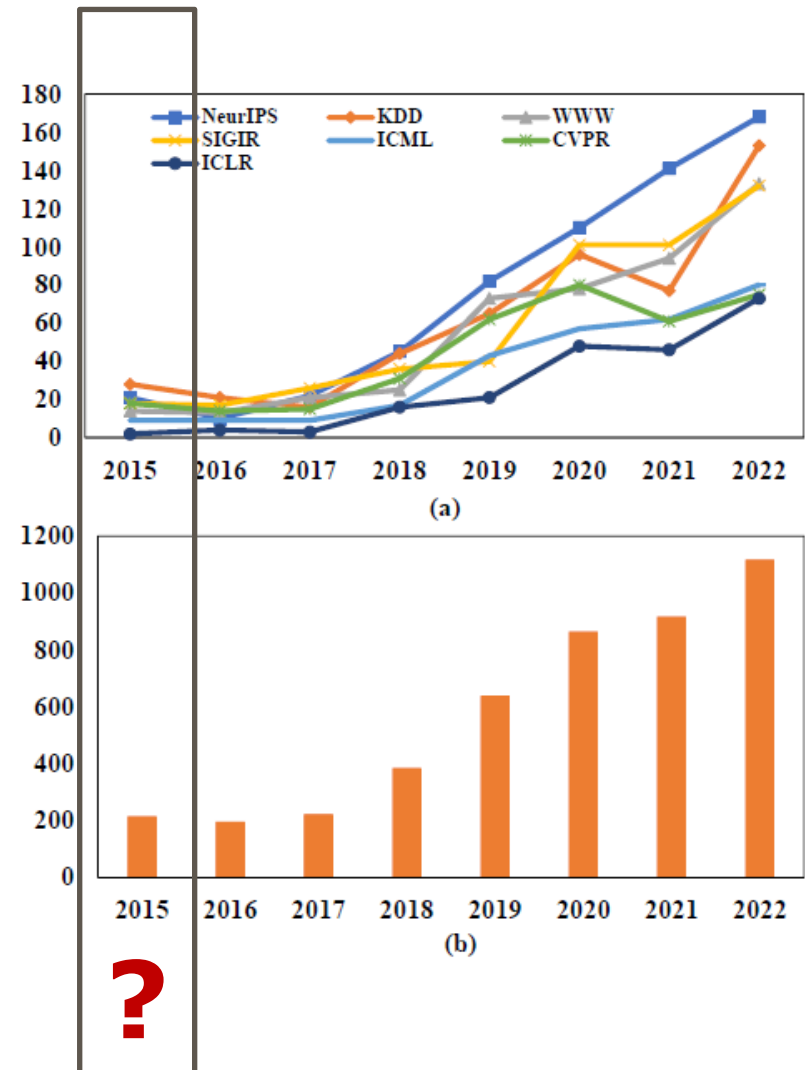


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- Not surprisingly, nowadays, a world-wide “exploding” area:
 - Research literature is rapidly growing
 - And concrete real-world applications as well

Number of GL publications in recent years **on top venues**, including NeurIPS, KDD, WWW, SIGIR, ICML, CVPR and ICLR. Figures (a) and (b) depict the number of GL publications at top conferences and the **total number**, respectively

- Including different terminologies for NN models, e.g.:
 - *Networks for Graphs / Graph Neural Networks (GNN) / Graph Convolutional networks (GCN)/ ...*
 - **Deep Graph Networks (DGN) framework** [Neural Netws. 2020]
 - ...



DGN: Origins

Pioneered in [Pisa](#) and [Siena](#), Italy,
2005-2009: **20th anniversary!**



1. **Recursive NN: GNN** [1], GraphESN [3] (see later)
2. **Layering/Convolutional: NN4Gs** [2] in constructive version, and then continued as **Graph Convolutional NN (GCN)**
 - A line of research attracting more and more world-wide research since 2015...

- [1] F. Scarselli, M. Gori, A.C.Tsoi, M. Hagenbuchner, G. Monfardini. **The graph neural network model**, *IEEE Transactions on Neural Networks*, 2009
- [2] A. Micheli. **Neural network for graphs: a contextual constructive approach**, *IEEE Transactions on Neural Networks*, 2009
- [3] C. Gallicchio, A. Micheli. **Graph Echo State Networks**, *IJCNN*, 2010
- [4] D.Bacciu, F. Errica, A. Micheli, M. Podda. A gentle introduction to deep learning for graphs, *Neural Netws.* 2020

IEEE TRANSACTIONS ON NEURAL NETWORKS

Neural Network for Graphs: A Contextual Constructive Approach

Alessio Micheli, Member, IEEE

Abstract—This paper presents a new approach for learning in structured domains (SDs) using a constructive neural network for graphs (NN4G). The new model allows the extension of the input domain for supervised neural networks to a general class of graphs including both acyclic/cyclic, directed/undirected labeled graphs. In particular, the model can realize adaptive contextual transactions, learning the mapping from graphs for both classification and regression tasks. In contrast to previous neural networks for structures that had a recursive dynamics, NN4G is based on a constructive feedforward architecture with state variables that use neurons with no feedback connections. The neurons are applied to the input graphs by a general traversal process that relaxes the constraints of previous approaches derived by the causality assumption over hierarchical input data. Moreover, the incremental approach eliminates the need to introduce cyclic dependencies in the definition of the system state variables. In the traversal process, the NN4G units exploit (local) contextual information of the graphs vertices. In spite of the simplicity of the approach, we show that, through the compositionality of the contextual information developed by the learning, the model can deal with contextual information that is incrementally extended according to the graphs topology. The effectiveness and the generality of the new approach are investigated by analyzing its theoretical properties and providing experimental results.

Index Terms—Cascade correlation, contextual transactions, graph patterns, learning in structured domains (SDs), neural networks for structured data, recursive neural networks.

I. INTRODUCTION

THE study of learning in structured domains (SDs) allows for a generalization of machine learning (ML) approaches to the treatment of complex data, offering both new impulses for theory and applications. In particular, extending the methodologies in order to deal with classes of data structures can be recognized as one of the most promising ways of increasing the possibility of successful new applications. Indeed, varying-size structures, e.g., sequences, trees, and graphs occur in many fields such as natural language processing (NLP), document and structured text analysis, image processing, object recognition, cheminformatics, and bioinformatics. Learning in SD raises theoretical issues considering first the need of new models for non-vectorial data, which must be able to directly represent relational data of variable size in their hypothesis space, up to the generalization of the ML basis induced by the extension of the

data domains. Indeed, various methodologies approach the topic starting from different paradigms in the machine learning, statistical relational learning (see, e.g., [1] and [2]), and relational data mining area (see, e.g., [3]), with the aim of extending the class of input data that are traditionally restricted to a flat-vector form. Often the terms "structured domain learning" (SDL) or "relational learning" are used to refer to the topic.

In this paper, we focus on subsymbolic supervised learning (classification/regression) for discrete data structures. We are recently witnessing an increasing attention in this area to the development of various kernel-based approaches for structured data (e.g., see [4]–[8] for overviews). Classical kernel approaches rely on the definition of a proper similarity measure on structures (expressed by the kernel function). An open issue is the problem of the design of a similarity measure on structures effective for specific tasks and/or of general validity, which poses practical and theoretical limitations [9], [10]. The point at issue is the concept of adaptivity, i.e., the capability that allows the model in principle to learn from the training data the similarity measure on structures for the task at hand. In this respect, neural network approaches based on adaptive processing of the structured input data constitute a powerful methodology in the area of subsymbolic approaches to SDL. In particular, advanced models such as recurrent neural networks for sequence domains and recursive neural networks (RNNs) for SDs have been introduced [11]–[13]. In the family of RNNs, constructive approaches have been introduced for sequences and structures as well, such as recurrent cascade correlation [14] and recursive cascade correlation (RCC) [12], [15], [16]. RNN models realize an adaptive processing (encoding) of recursive (hierarchical) data structures. A recursive traversal algorithm is used to process (encode) all the graph vertices, producing state variable values for each visited vertex.

The RNN approach is computationally characterized by the causality of the transition system [11]. The causality assumption imposes on a directed acyclic graph that the computation for a vertex depends only on the current vertex and vertices descending from it (see Section II-B). Such assumption is used by RNN (and RCC) to realize an encoding that matches the input hierarchical structures according to their topological order. However, such encoding process poses constraints on the transaction realized by the RNN models (see [17]) and on the classes of data. Indeed, sequences, rooted trees, or directed (positional) acyclic graphs (DAG¹ and DPAG) are subclasses of graphs for which a topological sort exists.²

¹Manuscript received May 02, 2007; revised January 19, 2008 and July 29, 2008; accepted August 18, 2008. First published February 6, 2009; current version published February 27, 2009.

²The author is with the Dipartimento di Informatica, Università di Pisa, 56127 Pisa, Italy.

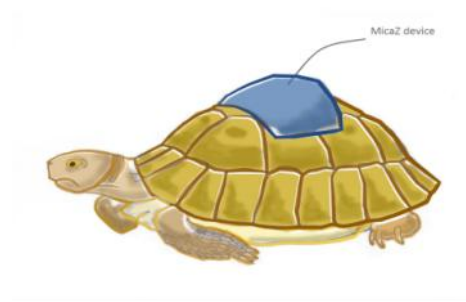
³This manuscript of DGN by RNN is introduced in [18].

Application Overview

@CIML



Chem/Bio-informatics



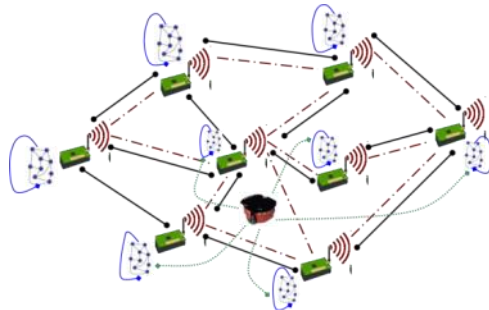
Pattern recognition on
animal activity



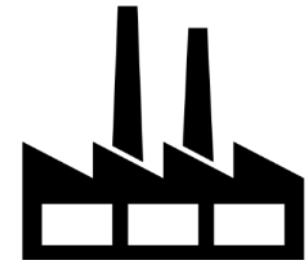
Sensor data
Smart *
IoT



Human Activity Recognition for
Health care (EU project)



Intelligent Sensor Networks (EU project)



Data analysis for
industrial leaders

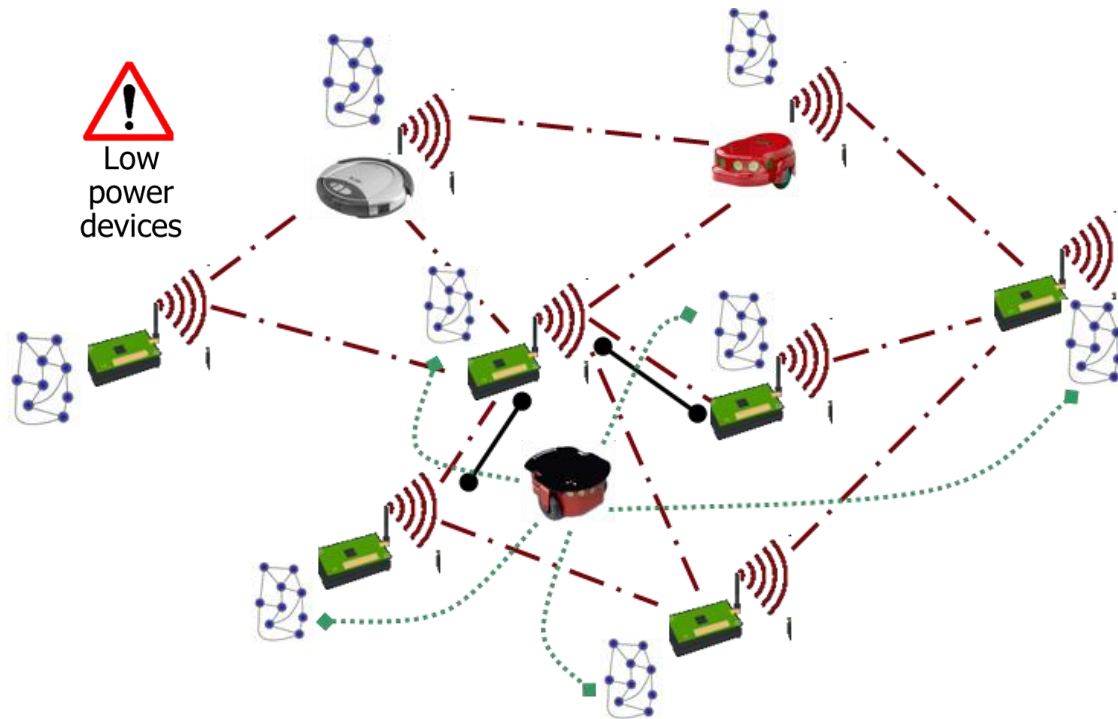
BACK 
TO **PAST**
THE

PAST PROJECTS
Examples

CIML – Zoom on an Application Example

Development of **basic and applied** research on Machine Learning

- EU Projects: FP7: Rubicon (self-learning robot ecologies)



- Distributed, cooperative, adaptive robotic ecology system
- WSN connections
- Machine learning modules (RNN) residing/embedded on ecology nodes/devices (on the edges)
- e.g. control of a robot / HAR / AAL / health monitoring, ...

Rubicon (EU FP7)



Dip. Informatica
University of Pisa

Reti robotiche per assistere gli anziani e i disabili



UNIVERSITÀ DI PISA

UNIPINews

Sito News Persone

ENGLISH 中文 ZHONGWEN

UNIPINews AREA GIURIDICO-ECONOMICA AREA MEDICA AREA SCIENTIFICO-TECNOLOGICA AREA UMANISTICA

Reti robotiche per assistere gli anziani e i disabili

L'Università di Pisa partner del progetto di ricerca europeo Rubicon finanziato con 2,5 milioni di euro



Si chiama RUBICON ed è un robot capace di dare assistenza ad anziani e disabili dopo averne imparato le abitudini di vita. Un team di ricercatori del dipartimento di Informatica dell'Università di Pisa, coordinati da Stefano Chessa e Alessio Micheli, è partner di un progetto di ricerca europeo finanziato con 2,5 milioni di euro che ha l'obiettivo di unire dispositivi robotici e reti di sensori in un sistema capace di interagire con l'ambiente circostante. Della ricerca, il cui capofila in Europa è l'Università di Dublino, fanno parte anche l'Istituto di scienza e tecnologia dell'informazione "A. Faedo" (ISTI) del CNR, la Fondazione Stella Maris, la Örebro University (Svezia), la University of Ulster (Nord Irlanda), due aziende spagnole di componenti robotici (Tecnalia e Robotnik) e l'azienda irlandese Pintail Ltd.

L'idea che sta alla base del progetto RUBICON (Robotic Ubiquitous Cognitive Network) è progettare robot in grado di apprendere le caratteristiche dell'ambiente e della qualità di vita della persona da assistere tramite una rete di sensori installati in casa. RUBICON è infatti un misto di "intelligenza" ambientale e robotica, una sorta di "cervello artificiale pervasivo" che fa parte dell'ambiente ed è capace di apprendere dall'ambiente stesso ciò che avviene e di comandare i robot di conseguenza. I sensori acquisiscono le informazioni, memorizzano un modello di abitudini ritagliato sulla persona specifica a cui portare assistenza e lo trasmettono

ai robot.

I ricercatori spiegano che, con questa tecnologia, sarà possibile abbinare un evento a un comportamento specifico del robot. Per fare un esempio, se RUBICON capisce che la persona ha l'abitudine di alzarsi tutte le mattine alle sette, il sistema attiva il robot a quell'ora per portare le pentole e assistere il soggetto specifico. Il vantaggio è che, se col tempo le abitudini cambiano, RUBICON è in grado di adattarsi autonomamente alle condizioni ambientali mutate, senza bisogno di essere riprogrammato.

Il team di ricerca dell'Università di Pisa, che include i gruppi di reti wireless e di intelligenza computazionale, è composto da 6 persone tra professori, ricercatori, assegnisti di ricerca e collaboratori, e ha lavorato negli ultimi sei mesi per definire il progetto che adesso passerà alla fase più operativa. RUBICON arriverà, infatti, a test di prototipi legati a scenari applicativi con una sperimentazione di sei mesi all'ospedale della Stella Maris e, parallelamente, in un living lab dell'azienda Tecnalia e Silbo.



Ne hanno parlato:

Ansa
Unità Toscana
Nazione Pisa
Corriere Fiorentino
Nuovo Corriere di Firenze
RepubblicaFirenze.it
CorriereFiorentino.it
TirrenoPisa.it
NazionePisa.it



Nel primo caso è previsto l'utilizzo del robot per il trasporto di materiale ospedaliero: i sensori ambientali dovranno essere addestrati a memorizzare dati utili a comprendere e disegnare il percorso per arrivare ai degenti. Il laboratorio spagnolo, invece, riprodurrà l'ambiente di un appartamento dove attori simuleranno azioni quotidiane sotto l'osservazione di sensori. Al suo lato sarà attrezzata una sala regia, dove i ricercatori potranno elaborare i dati, osservando i robot e la loro interazione con le persone.

Lo sviluppo di questa tecnologia apre nuove prospettive per il futuro: la sfida è riuscire a progettare sistemi che, in virtù della loro capacità di adattarsi alle persone, siano in grado di capire quello di cui hanno veramente bisogno, portando assistenza dove e quando serve, compatibilmente con la sensibilità del soggetto assistito, imparando a interagire con lui in maniera discreta e non invasiva.

ENGLISH NEWS

RASSEGNA STAMPA

- » Tirreno Pisa - Via le facoltà, ecco i dipartimenti
- » Repubblica Firenze - Pagine agli atenei, la ricerca sotto esame
- » La Stampa - Profumo: "Un nuovo patto per scuole, ricerca e imprese"

RASSEGNA VIDEO



EVENTI

- » Le pratiche della chimica: ieri e oggi
- » Job Meeting Pisa
- » La magistratura e la politica

COMUNICATI STAMPA

ATHENET

REDAZIONE



www.unipi.it

Robotic Ubiquitous Cognitive Ecology for Smart Homes.

Journal of Intelligent & Robotic Systems - Springer.

Volume 80(1), pp 57-81, Dec 2015.

A cognitive robotic ecology approach to self-configuring and evolving AAL systems.

Engineering Applications of Artificial Intelligence - Elsevier.

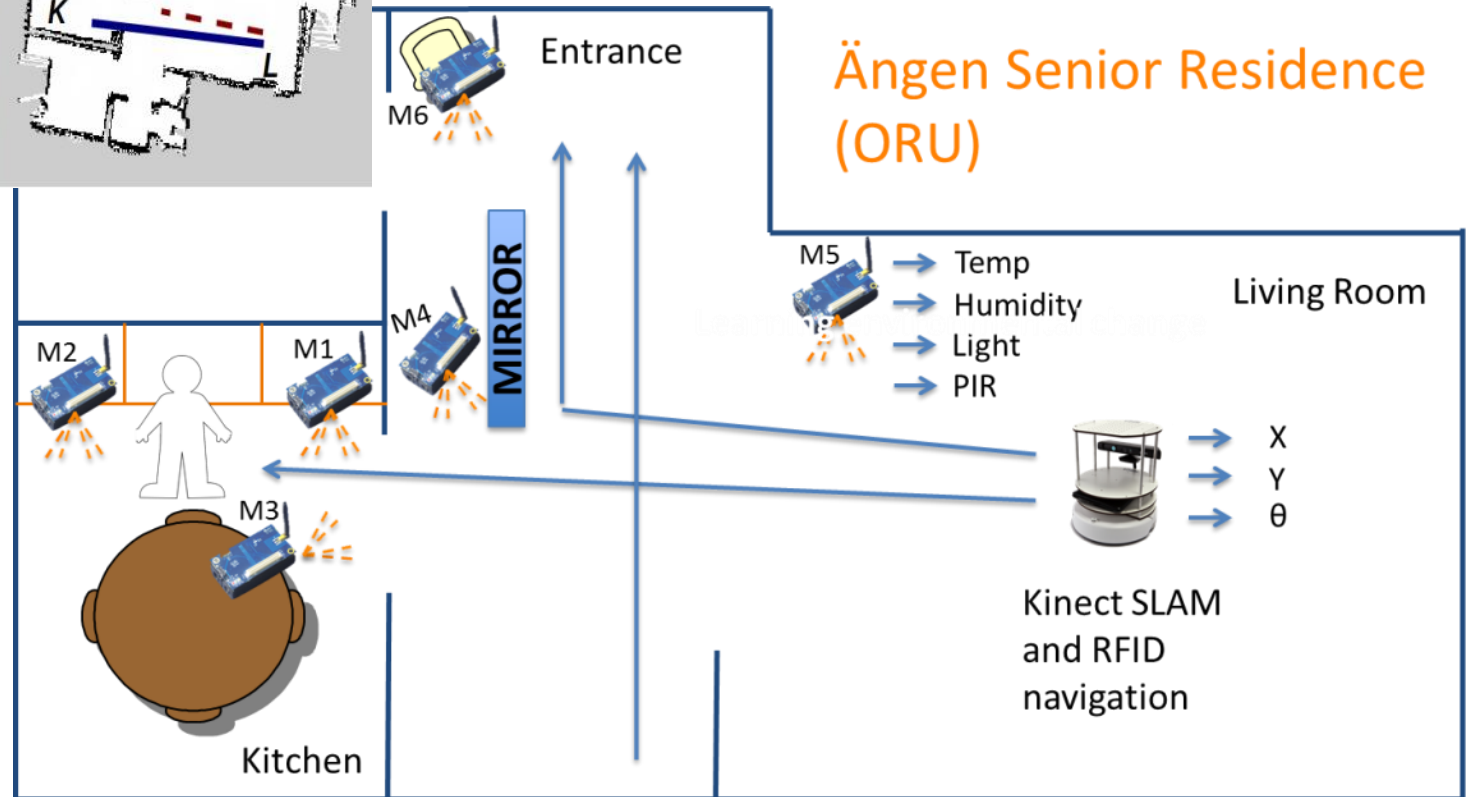
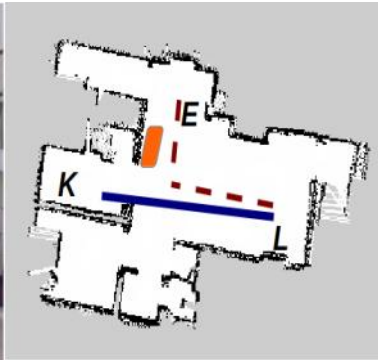
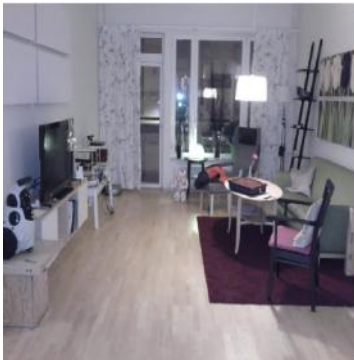
Vol. 45, October 2015, 269-280.

Examples of applicative scenarios: Robot Navigation



Robot navigation in Stella Maris Hospital
Using noisy signals from RSSI Anchors

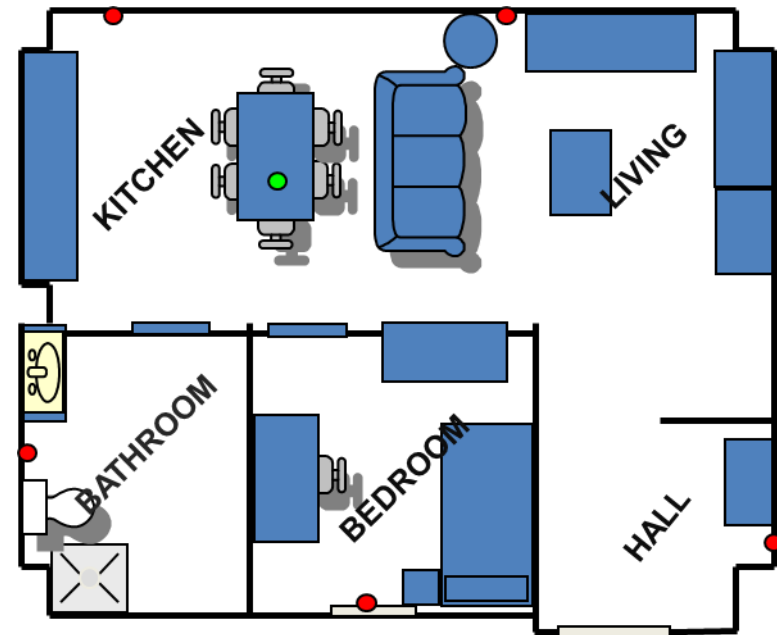
Examples of applicative scenarios: Learning for Robot Ecology



Learning/ adaptation to user's preferences and changes to the environment

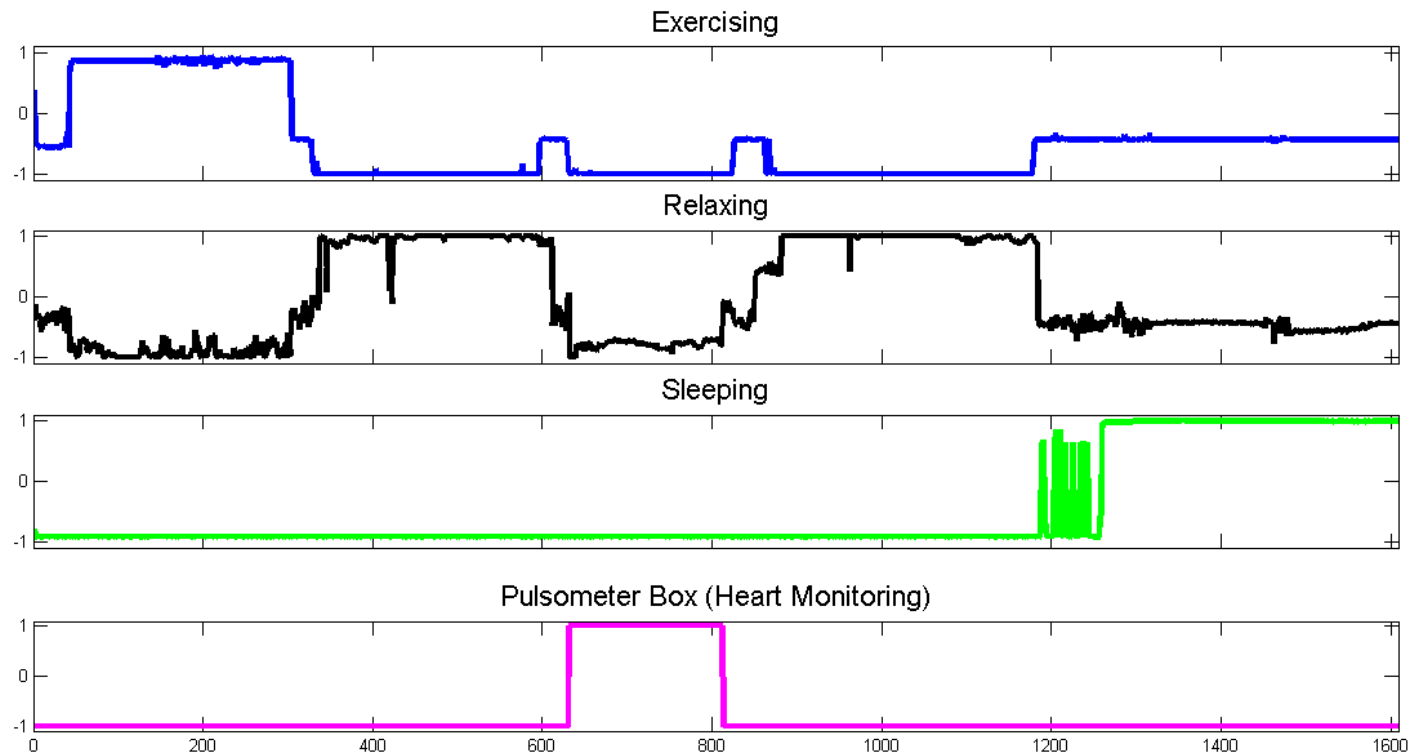
Examples of applicative scenarios: Ambient Assisted Living

- Predicting **event occurrence** and confidence of Human activities (from cooking to sleeping) basing on local sensors



AAL scenario at TECNALIA HomeLab
(Bilbao, Spain - 2014)

Human Activity Recognition

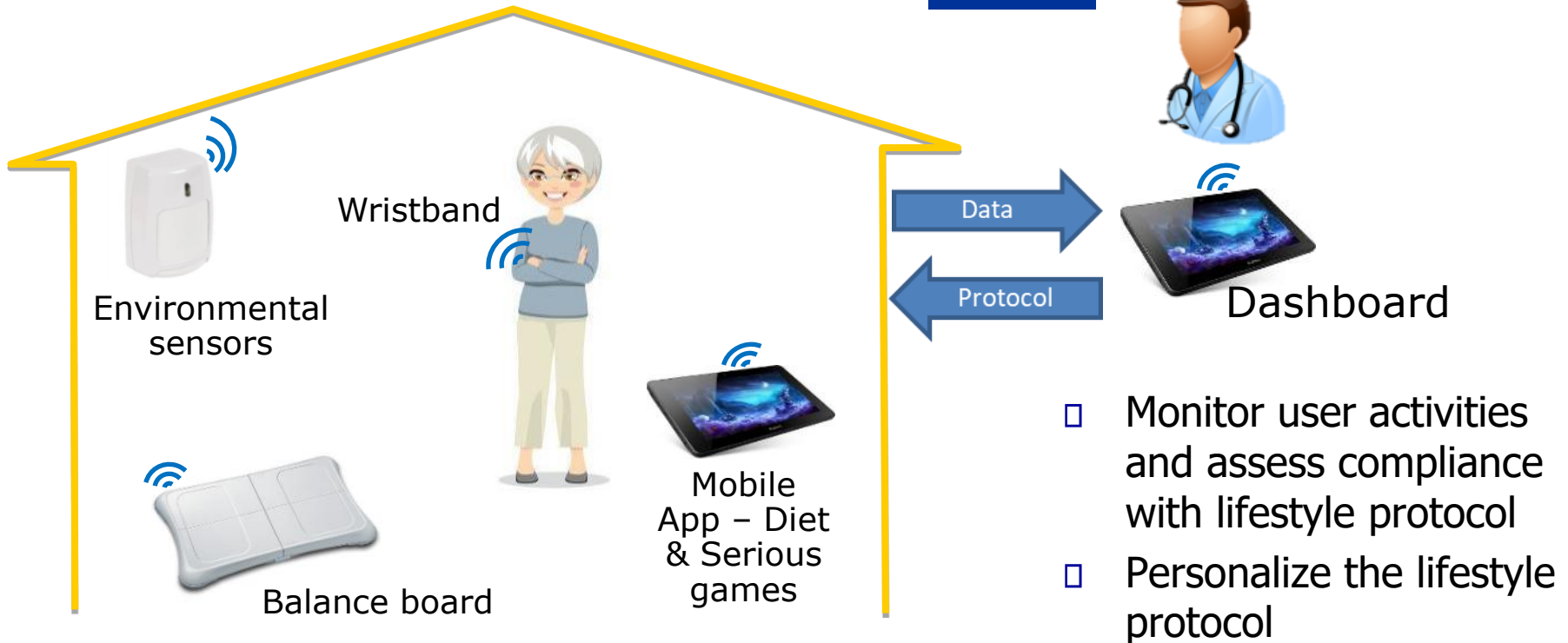


Inputs are signals from environmental sensors

Outputs of ESN Neural Networks (efficient models for temporal data)

EU FP7

Coordinated by IFC-CNR





DOREMI (EU FP7)



Dip. Informatica
University of Pisa

- Analysis of dynamical/temporal sensor data for predictive modeling of environmental context and user activities
- **Recognition of behavioral pattern for medical purposes, e.g.:**
 - Balance assessment
 - Level of physical activities/ Kcal energy consumption
 - Social interactions

from accelerometers and heart rate by a user bracelet and environmental sensors



DOREMI (start nov. 13): Decrease of cOgnitive decline, malnutRition and sedEntariness by elderly empowerment in lifestyle Management and social Inclusion



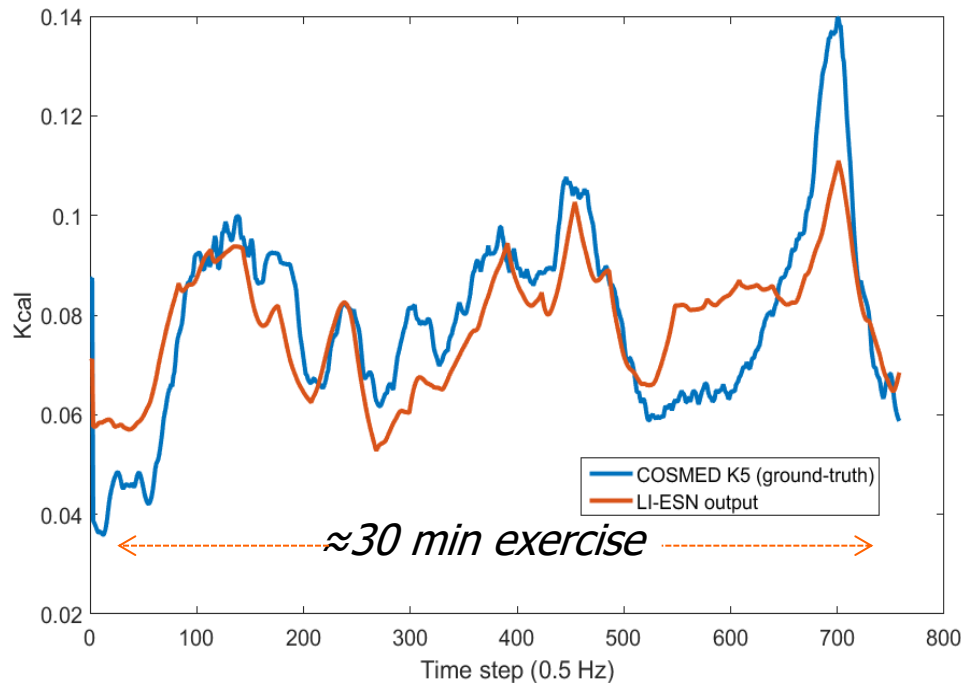


Physical Activity Expenditure Estimation

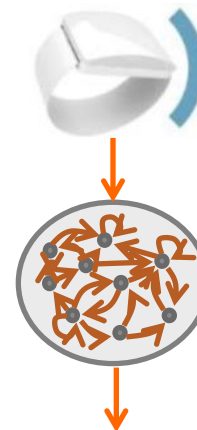


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- Automatic system for estimating calories consumption from heart rate (HR) measurements
 - During execution of the 22 DOREMI exercises
 - Ground-truth calories consumption (Kcal) from COSMED K5



- Improves literature approaches
- Tuned for elderly (by learning)
- Dramatically improves the accuracy of commercial devices (≈ 20 times better)



VS



Protezione Specie a Rischio

- Pattern recognition on tortoise activity (stream from sensor data) in natural parks (Unipi)



Tortoise@ recognizes the digging phase using neural networks

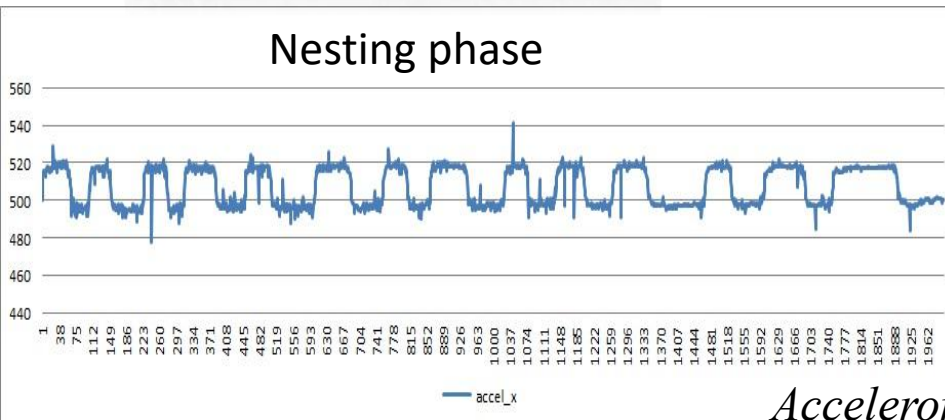


Eating phase

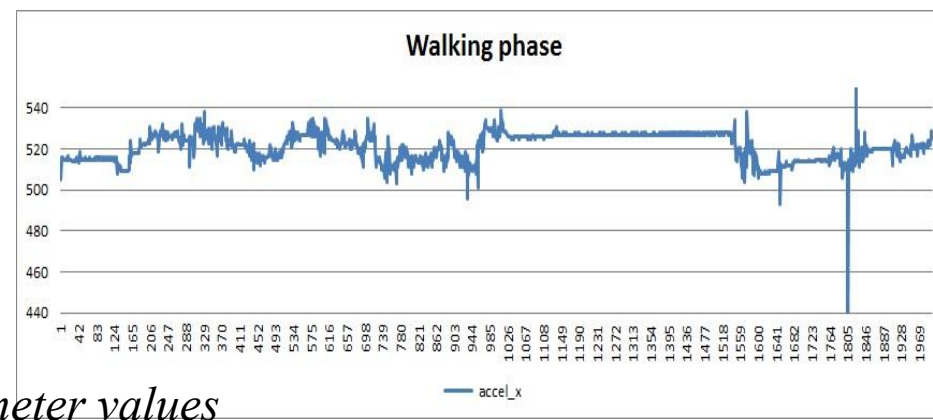


Walking phase

Nesting phase



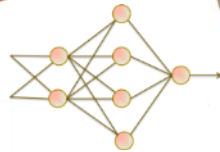
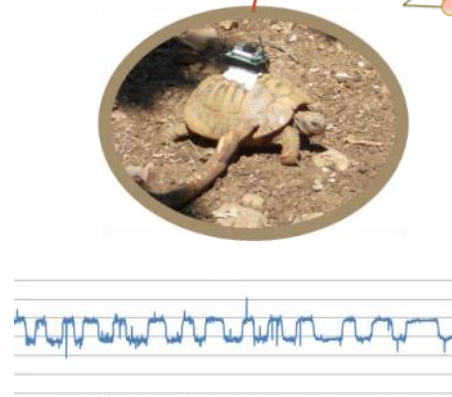
Walking phase



Accelerometer values

The final result ;-)

A device with a ML classifier identifies the excavation activity of the tortoises.



When the excavation phase is recognized, the device sends a message to the herpetologists.



The herpetologists take the eggs and protect them in a protection center.



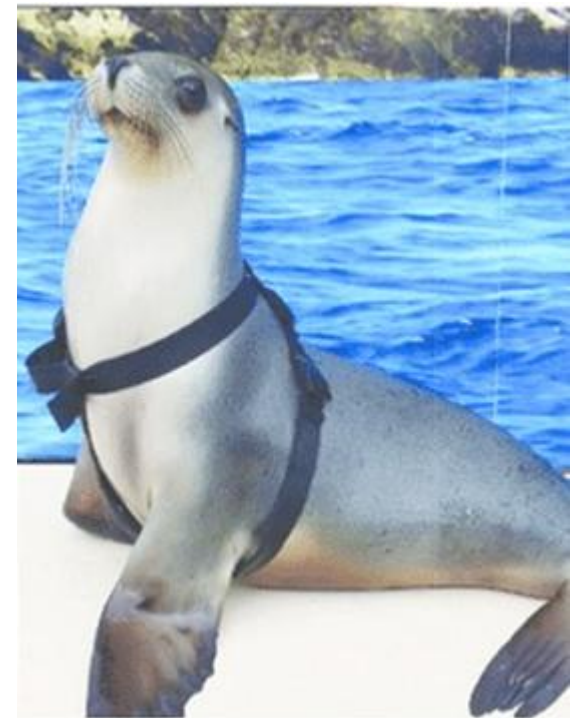
But also....



Penguins

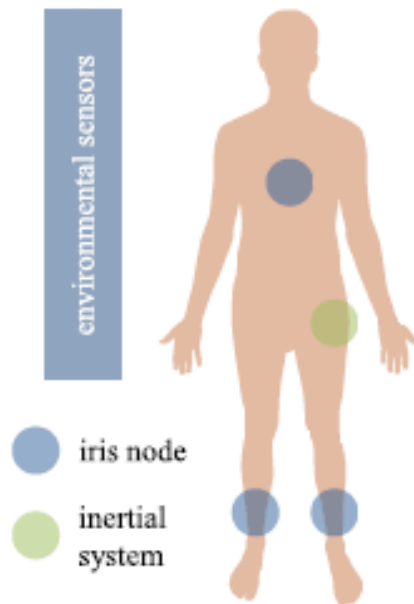
With Australian universities collaborations

Seals



And for Humans ...

- Human Activity Recognition by ESN



- Classification of human daily activities from RSS data generated by sensors worn by the user
 - Input: temporal sequences of RSS values (6 dimensional vector for each time step, noisy data)
 - Target: classification of human activity (bending, cycling , lying, sitting, standing, walking)
 - Extremely good accuracy ($\approx 0,99$) and F1 score ($\approx 0,96$)
 - 2nd Prize at **2013** EvAAL International Competition



Dataset is available online on the UCI repository

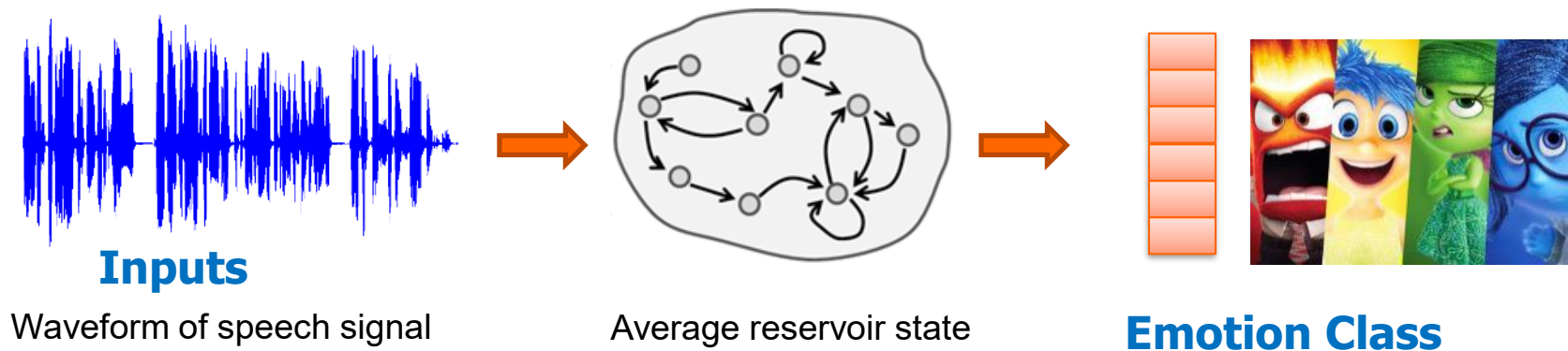
<http://archive.ics.uci.edu/ml/datasets/Activity+Recognition+system+based+on+Multisensor+data+fusion+%28AReM%29>

F. Palumbo, C. Gallicchio, R. Pucci, A. Micheli.

Human Activity Recognition using Multisensor Data Fusion based on Reservoir Computing.

Journal of Ambient Intelligence and Smart Environments. IOS Press, vol. 8, pp. 87-107, March 2016

Emotion Recognition from Speech by ESN



Speech Processing

EVALITA 2014 – Emotion recognition track (Sentiment Analysis)

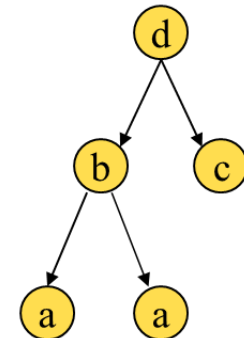
- **INPUT:** Time-series representing the waveform of speech directly (no explicit feature extraction required)
- **OUTPUT:** emotion (anger, disgust, fear, joy, sadness, surprise, and neutral)
- Efficiency and simplicity of the system
- Promising performances achieved

C. Gallicchio, A. Micheli. *A Preliminary Application of Echo State Networks to Emotion Recognition*.
Proceedings of CLiC-it & EVALITA 2014, December 2014.

Strings/Trees: ITA Emoji 2018

- Emoji Prediction or from strings or from trees
 - Strings: Italian tweets → Recurrent Neural Networks → Emojis class
 - Trees: Italian tweets (strings) → Parser → Tree (grammar relations and word embedding) → Tree Echo State Networks → Emojis class

				
20.27%	19.86%	9.45%	5.35%	5.13%
				
4.11%	3.54%	3.33%	2.80%	2.57%
				
2.18%	2.16%	2.03%	1.94%	1.78%
				
1.67%	1.55%	1.52%	1.49%	1.39%
				
1.37%	1.28%	1.12%	1.07%	1.06%



Emojis under consideration and their frequency within the dataset.

Medical/Health Applications



Neonatology

A **machine learning** approach to *estimating preterm infants survival*

- *Early* estimation *of* mortality *risk* of very preterm neonates
- From features such as birth-weight, gestational age, sex and other physiological variables (12) measured in the first minutes
- Allow to help doctor decision for immediate therapy, to better advise parents, etc.
- Data set of more than 28000 infant records (from the Italian Neonatal Network)
- **On-line** PISA (Preterm Infants Survival Assessment) predictor!
[<http://pisascore.itc.unipi.it/>]



A machine learning approach to estimating preterm infants survival: development of the Preterm Infants Survival Assessment (PISA) predictor
Scientific Reports - Nature (2018)
vol. 8:13743.



Unipi main page 16 Oct 2018



UNIVERSITÀ DI PISA

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ATENEOSTUDENTIRICERCAINTERNAZIONALETERRITORIO/IMPRESERPERSONALE



Il Monte Pisano tra i "Luoghi del Cuore"

L'Ateneo sostiene la candidatura per la campagna del FAI

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- » Matricolandosi
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- » Segreterie
- » Alice -Servizi online
- » Orientamento
- » Borse di studio e alloggio
- » Tirocini e Career Service
- » USID - servizi per disabili

RICERCA

- » Bandi ricerca
- » ARPI Archivio della Ricerca
- » Trasferimento tecnologico
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PERSONALE

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UNIPIEVENTI

- 16 Ottobre**

Thousands are Sailing
Mostra di foto d'epoca per il progetto Europea Migration
- 16 Ottobre**

Ludoteca Scientifica

UNIPINEWS



Dall'Intelligenza Artificiale uno strumento per valutare la sopravvivenza dei neonati prematuri

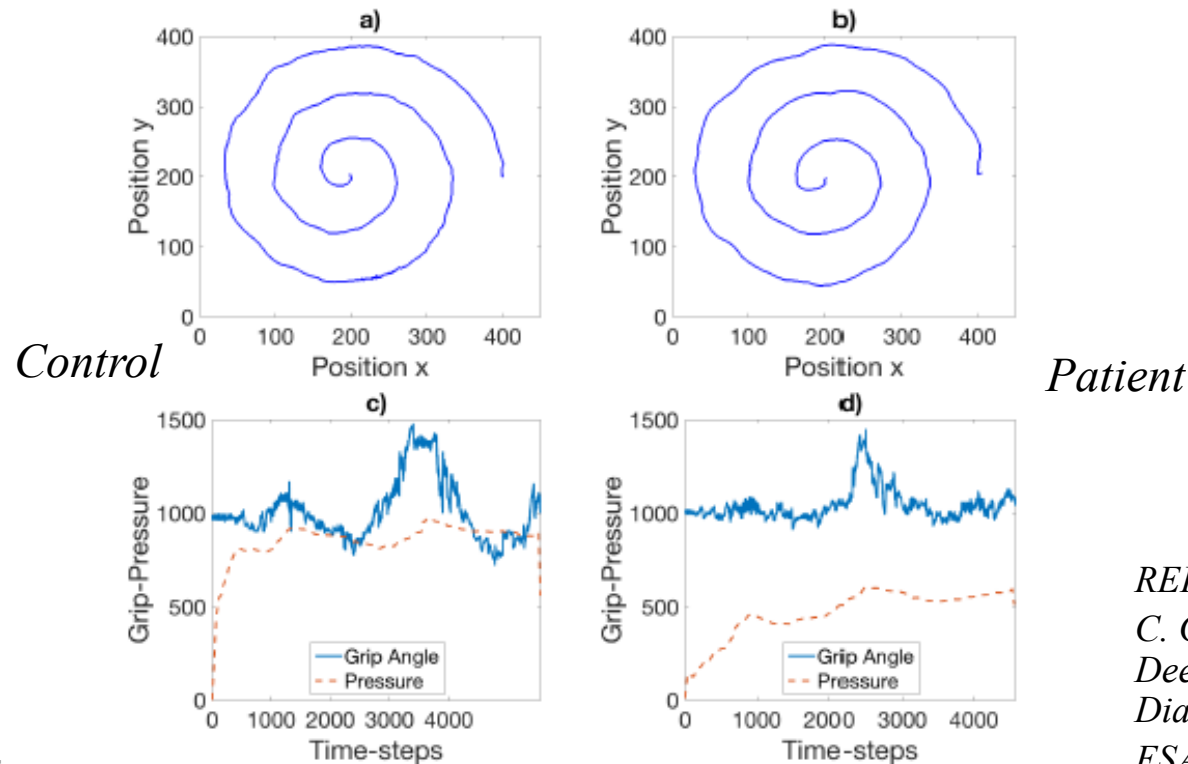
Si chiama PISA ed è stato realizzato dagli informatici dell'Ateneo e i neonatologi dell'Ospedale Versilia

[Vedi tutto \[English version\]](#)



Parkinson

- The identification of Parkinson diseases is performed by analyzing the **time-series** collected from a tablet device during the sketching of spiral tests
- by a **deep recurrent neural networks** (Deep ESN)



REF:

C. Gallicchio, A. Micheli, L. Pedrelli.
Deep Echo State Networks for
Diagnosis of Parkinson's Disease.
ESANN April 2018

Cardiology : Brugada

Brugada syndrome (BrS): not common but dangerous
(sudden cardiac death)

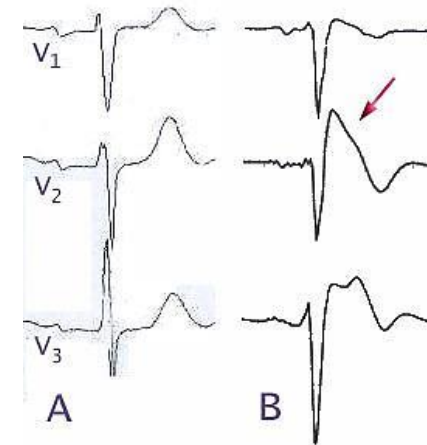
Freq. 5 per 10,000
Deaths 8% of sudden cardiac death

- No symptoms
- Brugada syndrome is diagnosed by identifying characteristic patterns on an electrocardiogram

- Input: ECG (electrocardiographic) signal
- Output: Brugada diagnosis alert

A: Normal electrocardiogram

B: examples of changes in Brugada syndrome (type B)



- A collection of labeled data progressively available by the project consortium



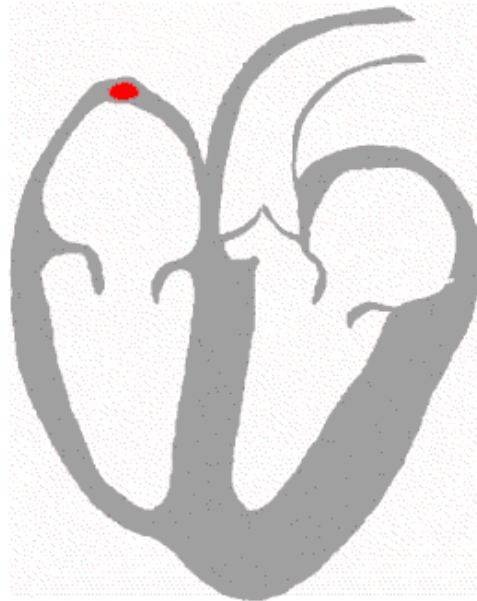
Regione Toscana

braid

**Thesis are
ready to start**

ECG

Electrocardiogram



Animation of a normal ECG wave:
cardiac cycle (heartbeat)

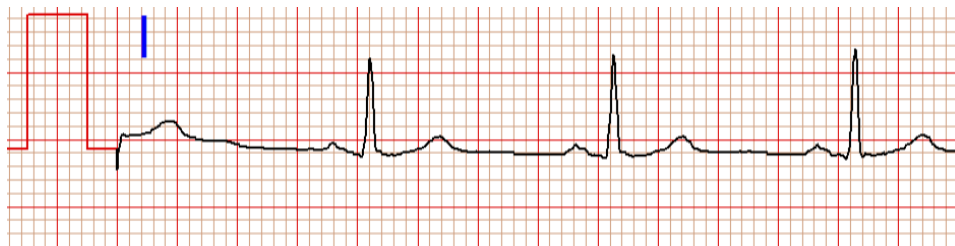
BrAID Project

Brugada syndrome and Artificial Intelligence applications to Diagnosis



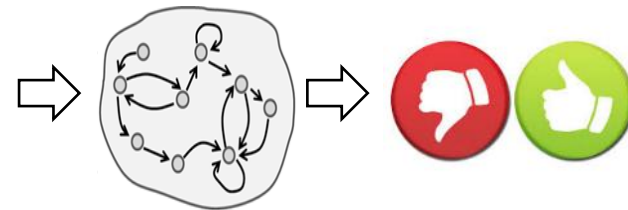
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- Design and to develop the ML models for the identification/detection of patterns related to the BrS diagnosis from the *ECG time-series*.
- Automatic (Brugada diagnosis alert), allowing advantages in term of *screening automatization*



ECG
(Electrocardiogram)

And *omics data*
(*biomarkers*)



RNN, ESN,
CNN, ...

Results (over 92%)
comparable or better
than clinicians



Regione Toscana

braid

Sensor data



Sensor data (I)

Sensor data from different sources:

- Data are noisy, calibration issues,
- typically no prior theory on treating them for prediction or to due to personalization, etc...
- IoT is going to provide large amount of data

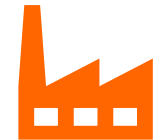
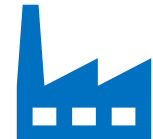
Hence well suited for ML approaches (see lecture 1, slide: *ML: When?*)

- Also, virtual sensor idea:
 - Calculate an estimate of a quantity of interest (a measure not available or very expensive to acquire) from available data such as known process parameters or low-cost existing sensors
 - Processes on production lines (manufacturing industry) or business processes, in finance, administration, clinical diagnosis, etc.

Sensor data (II)

We recently exploited this ideas for data

- From **mobile-phone** (classification of transport means) (PRJ with companies)
- From **public transport services** (PRJ with companies, [on-going](#))
- From **manufacturing industry** (lines and machinery for corrugated cardboard)
- From **seismic sensors** (PRJ)
- ...





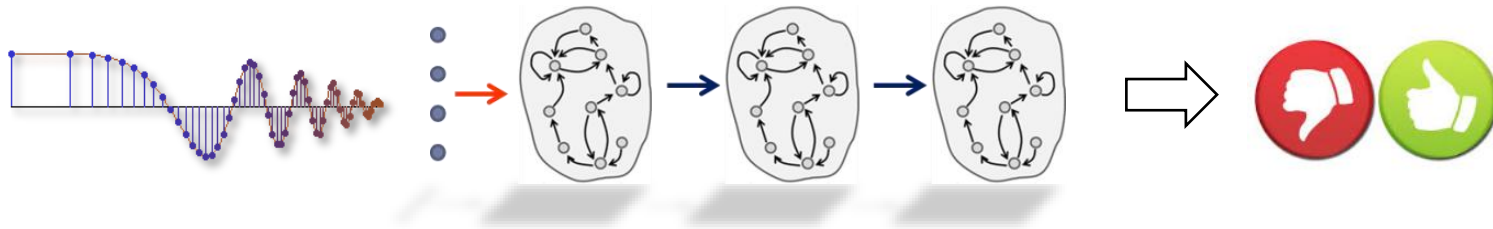
NEW PROJECTS (on going): and Examples of thesis proposals



New models (I)

Model proposal and analysis, but also software implementations ... examples:

- Efficient **Deep** Recurrent Neural Networks (DeepESN)



Gallicchio, Micheli, "Deep Echo State Network (DeepESN): A Brief Survey". ARXIV 2017, <https://arxiv.org/abs/1712.04323>

- Various architectures variants can be investigated starting from our available implementation (Python or Matlab code)
- Inventing new variants is our work! So, stay tuned !

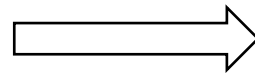
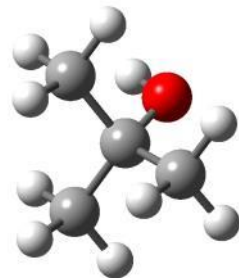


Thesis are possible

New models and theory (II)

Model proposal and analysis, but also software implementations ...
examples:

- Graphs** by Neural Networks



$$\left\{ \begin{array}{ll} +1 & \text{skull and crossbones} \\ -1 & \text{smiley face} \end{array} \right.$$

Property Value
e.g Toxicity:

Micheli 2009, Neural network for graphs: A contextual constructive approach. IEEE Transactions on Neural Networks, 20(3):498-511, 2009.

- With many [on-going projects](#) in the following
- Also: Collaborations with *Mario Negri Institute* of Pharmacological Research - Department of Environmental Health Science

PNRR Project NextGenerationEU



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- **FAIR** – Future Artificial Intelligence
Research is the Italian project approved in the call “Extended Partnerships: **Artificial Intelligence. Fundamental aspects**” of the PNRR (NextGenerationEU)
- 3 years starting on January 2023
- The FAIR project brings together >350 researchers and is developed in 10 spokes
- Advance frontier research in AI, both for fundamental and applied research development of *human-centric, sustainable, secure, inclusive and trustworthy AI* (explainability, robustness, privacy, fairness, ...)
 - Spoke 1: *Human-centered AI* (Leader: **UNIFI**), e.g.:
 - Objective 1: “human-in-the-loop” machine learning and reasoning
 - WP1.3 - Human centered ***Lifelong Learning for Complex Data***



FUTURE AI RESEARCH



New: Trustworthy for Graphs

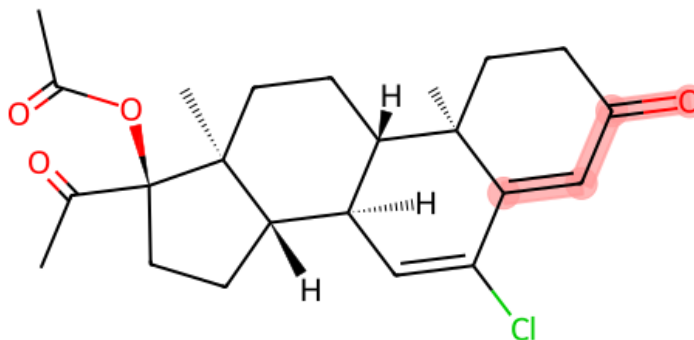


Trustworthy AI for complex data (structured domains):



- **Explainability**

- Interpretate the outcome of a NN for graphs, e.g. in the molecular domain, *why this molecule is active or is toxic?* Why this drug his effective?
- Or connecting to knowledge graphs, bridging symbolic and sub-symbolic reasoning, etc.



THE: PNRR Project NextGenerationEU



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- THE- TUSCANY HEALTH ECOSYSTEM
- *Ecosistema dell'innovazione sulle scienze e le tecnologie della vita in Toscana*
- 3 years starting on December 2022



- *Spoke: Precision medicine & personalized healthcare*
- System biology, AI and BigData for omics and computational biology
 - Investigation of machine learning for structured data in the omics and computational biology domain: *proteins, drugs and genomic structures.*

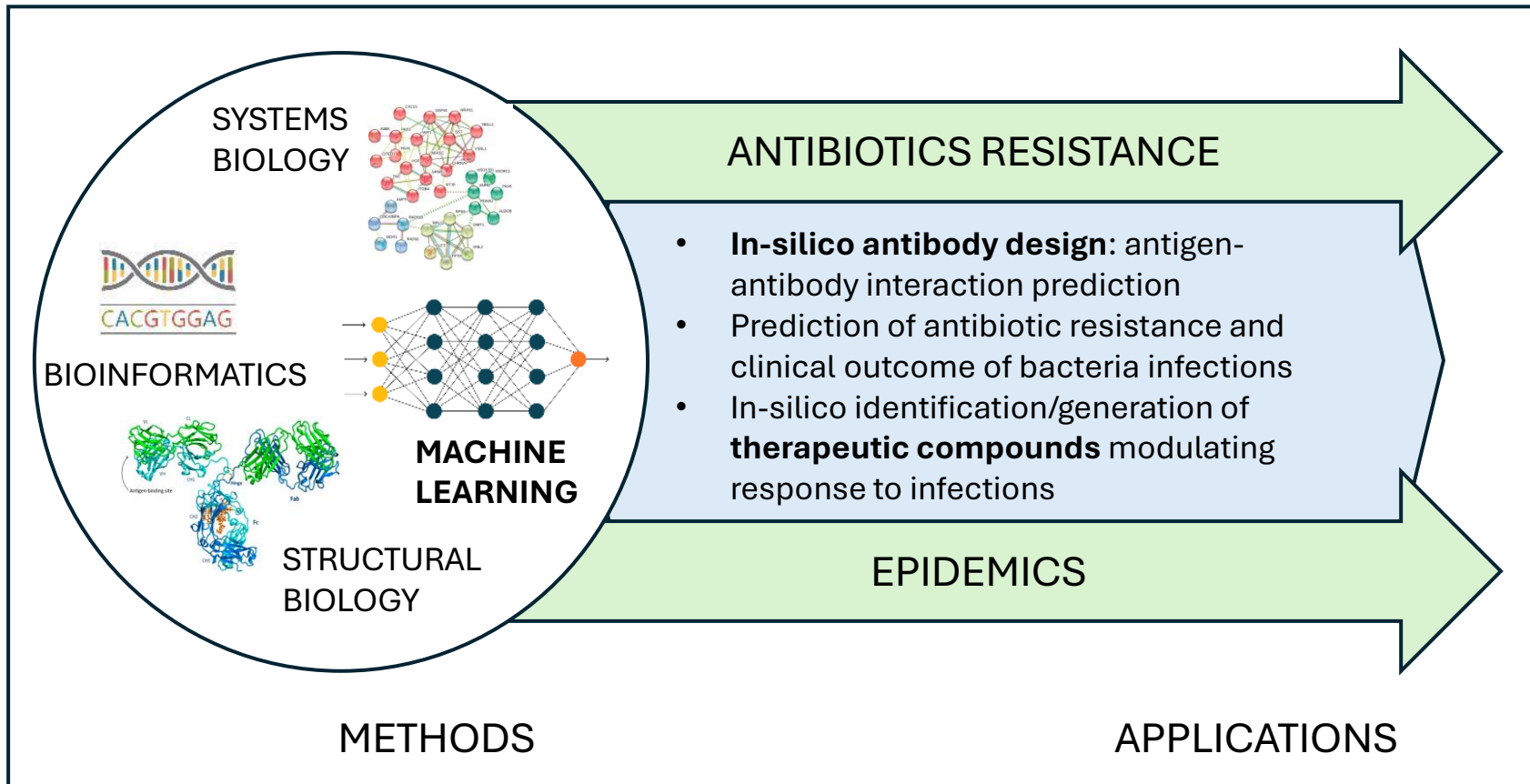


PAN-HUB



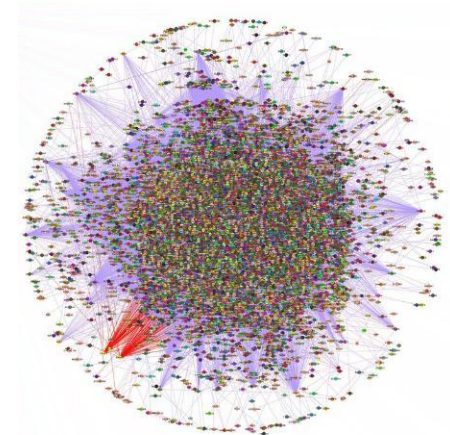
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Multidisciplinary and Interregional Research
and Clinical Trial Hub for Contrasting Pandemics
and Antimicrobial Resistance (PAN-HUB)



Ex.1: Bio-molecular Networks

- Protein-Protein Interaction Network (**PPI**)
 - Nodes represent proteins
 - Arcs represent physical associations between two proteins
- **Predicting sensitivity** (perturbation analysis) on PPI
- Starting from biochemical pathways (Biomodels) and mapping to PPI subgraphs (BioGRID, **60K nodes**)



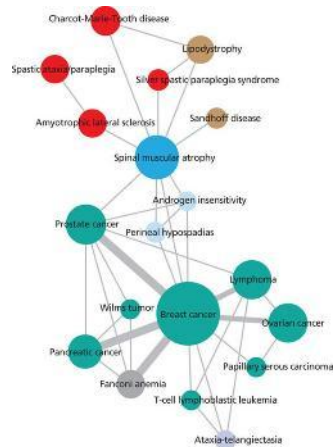
- **Results:**
 - **Efficiency** vs numerical simulation on biochemical pathways (ODE)
 - **High performance** even without protein embeddings, meaning that DGN is relying on PPI topology

PAN-HUB prj on
antibiotics resistance

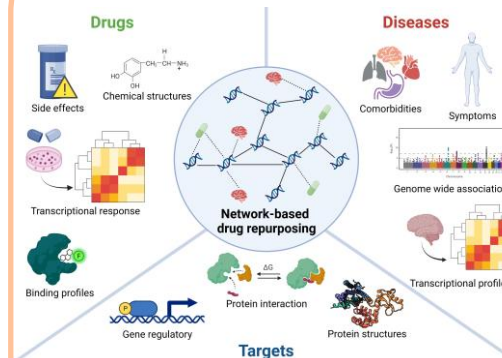
Dipalma, Fontanesi, Micheli, Milazzo, Podda. Sensitivity analysis on protein-protein interaction networks through deep graph networks *BMC Bioinformatics* (**2025**)



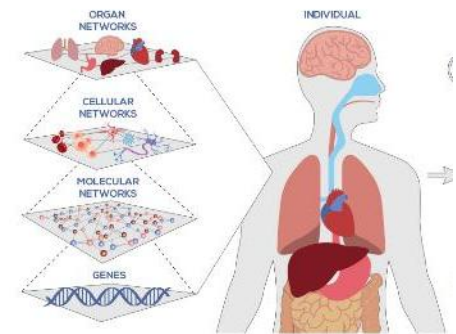
Impacts



**Disease
understanding**



**Drug design
and repurposing**



Precision medicine

Ex.1: Prediction of proteins function

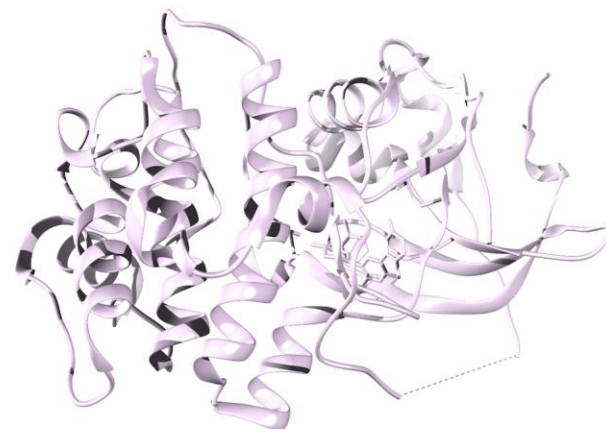


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- Prediction of proteins function
- Using gene ontology graphs
- Possibility to participate to CAFA challenge
 - There are many proteins in the databases for which the sequence is known, but the function is not.
 - The Critical Assessment of Functional Annotation (CAFA) is an ongoing, global, community-driven effort to evaluate and improve the computational annotation of protein function.

Approaches can include:

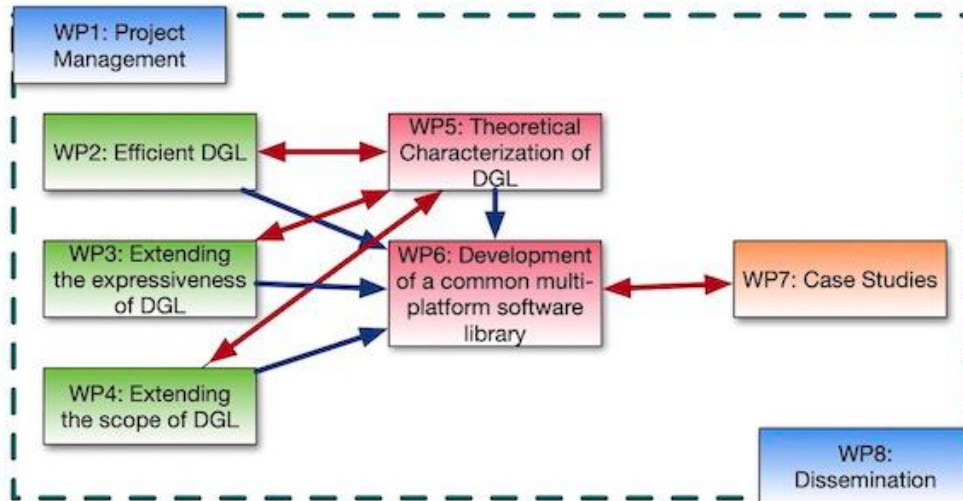
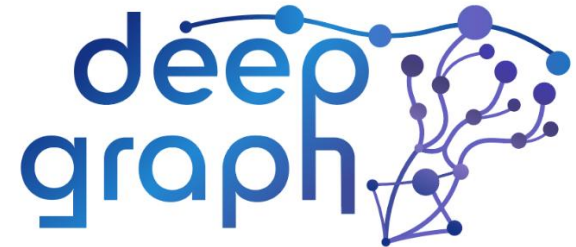
- SVM/**Kernel** building by efficient computation of common sub-graphs between pairs of proteins (with R. Grossi) [First thesis done]
- Use of NN for graphs



On going research: next project



- PRIN (national project), 2Y, **starting Feb. 2025**
- **DEEP-GRAPH: Design and Theory of Deep Graph Learning**
 - University of Padova (Prof. A. Sperduti)
 - University of Pisa (Prof. A. Micheli)
 - University of Siena (Prof. F. Scarselli)



Esempi di tesi (possibili)

Se tesisti motivati e volitivi (ML is hard!) ...



Esempi, 1to1 discussion:

- Speech recognition/audio detection
- Graph kernels & graph transformers
- Graph NN for Bio-networks
- Graph NN for Link prediction
- Network Quantification (node classification versus prediction of the class prevalence)
- ... Stay tuned!



Thesis are possible

Contact: micheli@di.unipi.it

Further projects - Companies

Increasing contacts by companies:

- Keep in touch for developments !!!

**Thesis
are
possible**



Bibliography

- For details on open challenges and projects ...
 - ❑ Use web sources (e.g. <https://ciml.di.unipi.it/>)
 - ❑ or ask to me (micheli@di.unipi.it)

;-)

Future – just ahead





DRAFT, please do not circulate!

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**Computational Intelligence &
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